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DESCRIPTION
INCLUSION OF NEW REFERENCES AND RELATED MODIFICATIONS

TES-K-100.01

Revision no. 01

**HEATING, VENTILATION AND AIR
CONDITIONING SYSTEMS DESIGN**

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TRANSMISSION ENGINEERING STANDARD

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1.0 GENERAL

1.1 Scope

This Transmission Engineering Standard prescribes the minimum mandatory requirements governing the design of Heating, Ventilating and Air Conditioning (HVAC) and Refrigeration Systems for National Grid Saudi Arabia facilities (i.e., building offices, control rooms, substations, communications, etc.).

This standard ultimately aims to metric ate all the units used. However, as most data, figures, charts, tables and other information available for calculations are still expressed in English units as it is felt that English units be also adopted as an interim for convenience.

The user of this Standard is advised to be conversant with HVAC terminology and applicable conversion units.

1.2 Conflicts and Deviations

Conflicts between this standard and other National Grid Saudi Arabia Engineering Standards, related Material Specification, Forms and Standard Drawings shall be resolved by the Manager, Standards and Specifications Department.

Any deviations providing less than the minimum requirements of this standard require written approval from the Manager, Standards and Specifications Department.

1.3 Industry Standards

The latest revision of the following Codes and Standards shall be applicable for the material covered in this TES.

AMCA 210	Laboratory Methods of Testing Fans for Certified Aerodynamic Performance Rating
ASME B31.5	Refrigeration Piping and Heat Transfer Components
ASME B31.9	Building Services Piping
AHRI-460	Performance Rating of Remote Mechanical-Draft Air-Cooled Refrigerant Condensers
AHRI -410	Forced-Circulation Air-Cooling and Air- Heating Coils
AHRI -430	Performance Rating of Central Station Air-Handling Units
AHRI -520	Performance Rating of Positive Displacement Condensing Units

ASHRAE Handbook Fundamentals

ASHRAE Handbook Systems and Equipment

ASHRAE Handbook Applications

ASHRAE Handbook Refrigeration

ASHRAE STD 62 Ventilation for Acceptable Indoor Air Quality

ASHRAE RP 185 Refrigerant Line Sizing

ASHRAE GRP 158 Cooling and Heating Load Calculation Manual

ANSI/ASHRAE

Standard 15 Safety Standard for Refrigeration Systems

Standard 16 Method of Testing for Rating Room Air Conditioners and
Packaged Terminal Air Conditioners

Standard 41.1 Standard Method for Temperature Measurement

ANSI/ASHRAE/

IESNA Standard 100 Energy Conservation in Existing Buildings

ASHRAE

Standard 41.2 Standard Methods for Laboratory Airflow Measurement

Guideline 20 Documenting HVAC&R Work Processes and Data Exchange
Requirements

Guideline 32 Sustainable, High-Performance Operations and Maintenance

Standard 189.1-Standard for the Design of High Performance Green
Buildings

Standard 55 Thermal Environmental Conditions for Human Occupancy

NEBB Procedural Standards for Testing, Adjusting and Balancing of
Environmental Systems

NEMA National Electrical Manufacturers Association

NFPA 90A Standard for the Installation of Air Conditioning and
Ventilating Systems

NFPA 101 Life Safety Code

NFPA 255	Standard Method of Test of Surface Burning Characteristics of Building Materials.
SMACNA	Duct Liner Application Standard
SMACNA	Architectural Sheet Metal Manual
SMACNA	HVAC Systems Duct Design
SMACNA	HVAC Duct Construction Standard
SMACNA	Industrial Ventilation Manual of Recommended Practice
UL 1995	Heating and Cooling Equipment
UL 555	Fire Dampers
UL 555S	Smoke Dampers
UL 984	Standard for Safety for Refrigerant Motor Compressors
UL 181	Factory-Made Air Ducts and Air Connectors
National Board of Fire Underwriters Pamphlet No. 46	
Accreditation Manual for hospitals	
ASTM A240	Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Pressure Vessels
ASTM A480	Specification for General Requirements for Flat Rolled Stainless Heat-Resisting Steel Plate, Sheet and Strip
ASTM A653	Specification for Steel Sheet, Zinc Coated (Galvanized) or Zinc-Iron Alloy-Coated by the Hot Dip Process
ASTM E41 - 92	Terminology Relating to Conditioning
ASTM E84	Test Method for Surface Burning Characteristics of Building Materials
ASTM E477	Test Method for Measuring Acoustical and Airflow Performance of Duct Liner Material and Prefabricated Silencers
ASTM E814	Test Method for Fire Tests of Through Penetration Fire Stops
AWS B2.2	Brazing Procedures and Performance Qualifications

	AWS D9.1	Sheet Metal Welding Code
	NAIMA AH124	Fibrous Glass Duct Liner Standard
1.4	Company Standards	
	01-TMSS-01	General Requirements for All Equipment/Materials
	73-TMSS-01	Air Conditioning Units, Packaged
	TES-H-107.02	Paint Color Codes and Standards
	TES-P-119.19	Substation Building and Site Development
	TES-P-119.21	Fire and Loss Prevention Requirements
	TES-T-111.18	Environmental Consideration For Communication Sites
1.5	Abbreviations	
	AABC	Associated Air Balance Council
	AMCA	Air Movement and Control Association
	ANSI	American National Standards Institute
	ARI	Air-Conditioning and Refrigeration Institute
	ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
	ASME	American Society of Mechanical Engineers
	NEBB	National Environmental Balancing Bureau
	NFPA	National Fire Protection Association
	SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
	UBC	Uniform Building Code
	UL	Underwriter's Laboratory
	SBC	Saudi Building Code
	UMC	Uniform Mechanical Code

ASTM	American Society for Testing and Materials
AWS	American Welding Society
NAIMA	North American Insulation Manufactures Association
AIHA	American Industrial Hygiene Association

2.0 OUTDOOR AMBIENT CONDITIONS

Temperatures given in this section shall be used as the ambient design temperatures for HVAC System design, equipment selection and equipment specifications purposes for the geographic areas listed. Design data for other locations, not given shall be determined by the Project Design Team and concurred by the Manager, Standards and Specifications Department.

2.1 Average Outdoor Design Condition in Saudi Arabia.

The following information shown on Table 1.0 is the average outdoor design condition for selected areas/locations as obtained from Meteorology & Environmental Protection Administration, Ministry of Defense & Aviation, Kingdom of Saudi Arabia.

2.2 Ambient Air Quality

HVAC Systems shall be designed for air-borne dust concentration of 1 mg/cubic meter (0.000001 oz/ft³ sand storm dust concentrations reaching 500 mg/m³ (0.0005 oz/ft³) near the ground. Dust particle sizes are:

- 95% less than 20 micrometers (0.0008 in.)
- 50% less than 1.5 micrometers (0.00006 in.)

Elements present in dust include compounds of sodium, calcium, magnesium, silicon and aluminum. When wetted or exposed to condensing dew point conditions these compounds become electrolytes and severely corrode unprotected metal surfaces.

Control and electrical panels in non-conditioned spaces shall be enclosed in airtight enclosures.

HVAC Systems and equipment for offshore and located within 1 km. from sea shall also be protected against failure and corrosion due to wind borne sea water spray and accumulation of salt.

Table 1.0 - Outdoor Ambient Design Conditions

Location	Summer			Winter	Wind	
	Design db, °C (°F)	Design wb, °C (°F)	Daily Range °C (°F)	Design db, °C (°F)	Mean Speed KPH	Direction
Abha	32(90)	24(75)	15(27)	3.2(38)	15	SW
Al-Baha	39(102)	24(75)	13.4(24.1)	5(41)	19	SW
Al-Hasa	50(122)	25((77)	16.8(30.2)	6.4(44)	11	N
Al-Jouf	47(117)	24(75)	16.8(30.2)	0.9(34)	20	W
Al-Qaseem	49(120)	24(75)	18.5(33.3)	3(37)	17	NE
Arar	48(118)	24(75)	17(30.6)	-2.5(27)	19	NW
Dhahran	49(120)	28(82)	14(25.2)	6(43)	22	N
Jeddah	48(118)	27(81)	14(25.2)	14(57)	22	N
Jizan	44(111)	28(82)	8(14.4)	7(45)	13	W
Hafar Al-Batin	50(122)	24(75)	17.7(31.9)	1.9(35)	19	NE
Hail	44(111)	24(75)	16.3(29.3)	-1.5(29)	17	NE
Khamis Mushait	35(95)	24(75)	15(27)	4.1(39)	15	SW
Madinah	48(118)	24(75)	17(30.6)	9(48)	15	W
Makkah	49(120)	26(79)	15(27)	15(59)	9	N
Najran	42(108)	24(75)	16(28.8)	6(43)	7.4	E
Nariyah	48(118)	30(86)	18(32.4)	7.5(45)	24	N
Riyadh	48(118)	24(75)	15.7(28.3)	5.1(41)	17	NE
Tabuk	45(113)	24(75)	15.6(28.1)	0.5(33)	15	N
Yanbu	48(118)	28(82)	14(25.2)	10.6(51)	19	W

3.0 INDOOR DESIGN CONDITION

This Section contains the minimum indoor design criteria for air conditioning in National Grid Saudi Arabia installations in Saudi Arabia.

3.1 Design Conditions for Special Occupancies

Design conditions for special occupancies shall, as a minimum, comply with recommendations of the ASHRAE Fundamentals, Systems, and Applications Handbooks; applicable health, safety, and construction codes and equipment manufacturers requirements.

3.2 Dry Bulb and Humidity

HVAC System shall be capable of maintaining dry bulb and relative humidity within the given performance range under climatic conditions of Section 2.0.

3.3 Indoor Conditions for Calculation

The conditions listed herein shall be used as indoor design conditions for cooling load calculations.

Table 2.0 - Indoor Design Condition

	FACILITY	DESIGN db °C (°F)	DESIGN wb°C (°F)	% RELATIVE HUMIDITY	AIR MOTION mps (FPM)	TOLERANCE db (rh)
1.	Offices, Schools, Theaters*	24 (75)	17 (63)	50	0.075 - 0.25 (15 - 50)	± 2°C (± 20%)
2.	Shops, Houses, Apartments, Trailers, Dining Halls, Stores*	24 (75)	17 (63)	50	0.075 - 0.25 (15 - 50)	± 2°C (± 25%)
3.	Computer Rooms***	22 (72)	17 (63)	50	0.075 - 0.25 (15 - 50)	± 2°C (± 20%) ± 2°C (± 20%)
4.	Comm. Facilities, Central Offices	See TES-T-111.18 Environmental Consideration For Communication Sites				
5.	Power Plants, BSP/ Grid Substations, Control Rooms****	25 (77)	17.8 (64.2)	50		
6.	Elect. Equipment Rooms:(Switchgear, MCC, Relay Test, Battery Room, etc.), Toilet****	25 (77)	17.8 (64.2)	50		

Table 2.0 - Indoor Design Condition (Cont'd)

	FACILITY	DESIGN db °C (°F)	DESIGN wb°C (°F)	% RELATIVE HUMIDITY	AIR MOTION mps (FPM)	TOLERANCE db (rh)
7.	Cable Entry/Cable Basement****	30 (86)	17.8 (64.2)			
8.	Hospitals & Clinics Various Rooms**	24 (75)		50		
9.	Laboratories	Refer to National Grid Saudi Arabia Hygienist				
10.	Masjid (Mosques)*	24 (75)	17 (63)	50		

* Refer to ANSI/ASHRAE Standard 55

** Additional design information is required

*** Computer room temperature shall be determined by equipment Manufacturer's requirement.

**** Refer to 01-TMSS-01

4.0 COOLING LOAD CALCULATIONS

4.1 Introduction

The underlying sciences of HVAC Engineering, such as thermodynamics and hydraulics are not discussed here. Principles derived from these sciences are used only in the context of how their use facilitates the design of HVAC Systems.

The method of load calculations and estimate sheet follow ASHRAE GRP 158. Figures, constants and other factors shall be taken from ASHRAE GRP 158 or from manufacturer's data.

4.2 HVAC Load Calculations

Typically, the following steps are involved in HVAC load calculations:

First, Data shall be gathered from building plans and specifications. Building location, orientation, shading characteristics, building dimensions, type of building materials and internal load profiles are all required for the subsequent calculations. Secondly, selection of indoor and outdoor design conditions. These data for the various National Grid Saudi Arabia locations in Saudi Arabia are listed in Sections 2.0 and 3.0. After that, the Calculation of heat transfer coefficients, calculation of space cooling load, calculation of space heating load and Psychrometric analysis shall be performed.

There are several computerized programs available in the market for HVAC calculations. These programs are based on the same data and methodology as the manual calculations. User of the computer programs should be able to translate calculation results to National Grid Saudi Arabia Standard Form No. 16305/07, Cooling and Heating Load Estimate Sheet. It is strongly recommended that the user of any HVAC computer software be familiar with the abbreviated procedure of calculations as outlined in the following sections of this document.

Overall building cooling and heating loads shall be calculated. Zoning, exposure and building mass shall be considered in the calculations. The Air Conditioning Systems shall provide comfort conditions in all spaces throughout the operating period. Each zone room and portion of room with different load profile orientation or sensible load shall be calculated. A separate block load for each air-handling unit shall be calculated.

During cooling load calculations, all sensible and latent heat sources shall be considered.

Sensible cooling load shall be calculated for building envelope, people, lights, equipment and for outside air that is introduced into the system by air make-up or by infiltration. Latent cooling load shall be calculated for people, outside air and any process in which moisture is given up to the air.

Heating loads shall include heat losses from outside surfaces (roofs, walls, windows, raised floors); interior non-conditioned spaces (partitions, ceilings, floors); make up air and infiltration; duct and plenum losses.

A Psychrometric analysis shall be provided for each air handler load. This analysis shall include, but not necessarily be limited to mixing temperature of outside and return air streams, coil conditions i.e., entering and leaving conditions, room design conditions and outside design condition.

4.2.1 Cooling Load

Space cooling load is the rate of heat removal from the conditioned space that has to be maintained in order to provide the desired inside conditions.

Cooling loads can be classified according to the heat gain mode of the conditioned space as follows:

External cooling load is as a result of heat gain through the roof, walls and windows of the building.

Internal cooling load has to compensate for the heat generated within the conditioned space by occupants, lights and equipment.

Ventilation and infiltration cooling load due to the heat gain from outside air which enters the building.

The use of Cooling and Heating Load Estimate Sheet (National Grid Saudi Arabia Form No.16305/07) is described as follows:

- a. Design Conditions
 - i. Outdoor design conditions shall be obtained from Section 2.0 of this standard.
 - ii. Indoor design conditions shall be obtained from Section 3.0 of this standard
 - iii. Latitude of specific projects shall be obtained from the geographic map of Arabian Peninsula
 - iv. Design month shall be August and design hour shall be 4:00 PM (1600 Hr).
 - v. Daily Range shall be obtained from Section 2.0 of this standard. Daily Range equals maximum outdoor temperature minus minimum outdoor temperature.

b. External Cooling Load (Heat Flow from Outside)

i. Conduction

- Roof, Wall, Glass
Use National Grid Saudi Arabia Form No.16307/07 to calculate CORRECTED COOLING LOAD TEMPERATURE DIFFERENCE (Area shall be obtained from Architectural Plans)
- Partition Wall, Ceiling, Floor
 Δt = Air temperature difference between the adjacent space and conditioned space. (Area shall be obtained from Architectural Plans.)

ii. Solar Radiation

- Identify the direction of glass from building orientation. See Architectural Plans.
- Shade Coefficient (SC). See ASHRAE GRP 158 or ASHRAE Fundamentals
- Solar Heat Gain Factor (SHGF) for the month, latitude and orientation
- No External Shading - See ASHRAE GRP 158.
- External Shading - See ASHRAE GRP 158
- Net area of glass shall be obtained from Architectural Plans.
- Cooling Load Factor (CLF) for glass:
- No Interior Shading - See ASHRAE GRP 158 or ASHRAE Fundamentals.
- Interior Shading - See ASHRAE GRP 158 or ASHRAE Fundamentals.

c. Internal Cooling Load

i. Lights

- Identify total wattage from Electrical Plans
- Usage (%) = $\frac{\text{Wattage in use}}{\text{Total Installed Wattage}} \times 100$
- Ballast Factor for Fluorescent Lights. See ASHRAE GRP 158. Ballast Factor for incandescent light is one (1).
- Cooling Load Factors (CLF). See ASHRAE GRP 158, or ASHRAE Fundamentals.

ii. People

- The number of people in the space shall be obtained from survey or ASHRAE GRP 158.
- Sensible Heat Gain (q_s) per person. See ASHRAE GRP 158 or ASHRAE Fundamentals.
- Sensible Heat Cooling Load Factors (CLF) for People, See ASHRAE GRP 158 or ASHRAE Fundamentals.

iii. Appliances

- Sensible Heat Gain from Appliances and Laboratory Equipment (q_s). See ASHRAE GRP 158 or ASHRAE Fundamentals.
- Sensible Heat Cooling Load Factor (CLF). See ASHRAE GRP 158 or ASHRAE Fundamentals.

iv. Equipment

Calculate additional heat gains due to heat emitting equipment using applicable equations or vendor's data.

d. Room Sensible Subtotal

e. Factor of Safety

A 10% safety factor added to the room sensible heat subtotal is considered strictly as a factor or probable error in the survey or estimate.

f. Room Sensible

g. Ventilation Load Due to Fresh Outdoor Air Make-Up

- i. Buildings for habitation are usually pressurized by mechanical ventilation by providing substantially more outdoor air than exhaust. Introduction of outdoor air for ventilation of conditioned spaces is necessary to dilute the odors given off by people, smoking and other internal air contaminants.
- ii. Ventilation Standards are available from any of the following:
 - ASHRAE GRP 158
 - ASHRAE Fundamentals

iii. Sensible Heat Gained, $q_s = 1.1 \times \text{CFM}_{\text{oa}} \times (t_{\text{oa}} - t_{\text{rm}})$, Btu/hr

where: CFM_{oa} - outdoor air flow

t_{oa} outdoor air dry bulb temperature, °F

t_{rm} room dry bulb temperature, °F

h. Latent Load

i. People

- The Latent Heat Gains (q_l) shall be obtained from ASHRAE GRP 158 or ASHRAE Fundamentals.
- The number of people in space shall be obtained from survey or ASHRAE GRP 158.

ii. Equipment

- The number and type of equipment shall be obtained from survey or Architectural Plans.
- The latent heat load (q_l) shall be obtained from ASHRAE GRP 158.

iii. Room Latent

iv. Ventilation Load

Latent Heat Gain, q_l (Item No. 35) = $0.7 \times \text{CFM}_{\text{oa}} \times (W_{\text{oa}} - W_{\text{rm}})$, Btu/hr

Where,

Humidity Ratio Difference between outdoor air (W_{oa}) and room air (W_{rm}) in Grains Moisture per lb. Dry Air that are obtainable from Psychrometric Chart.

i. Cooling Load Summary

- Room Sensible (Item No. 36) = Item No. 29
- Ventilation Air Sensible (Item No. 37) = Item No. 30
- Fan Motor Heat Gain (Item No. 38): See Table 4.12 of ASHRAE GRP 158
- Grand Sensible (Item No.39) = Item Nos. (36 +37 +38)
- Room Latent (Item No. 40) = Item No. 34
- Ventilation Air Latent (Item No. 41) = Item No. 35
- Grand Latent (Item No. 42) = Item Nos. (40 + 41)

- j. CFM Supply Air is the air quantity required to satisfy the room load.

$$CFM_{sa} = \frac{\text{Room Sensible Heat}}{1.10 (t_{rm} - t_{sa})}$$

where: CFM_{sa} - supply air flow

t_{sa} - supply air dry bulb temperature, °F

t_{rm} - room dry bulb temperature, °F

In determining the required air quantity, when neglecting the supplementary loads (Fan Heat Gain, Duct Heat Gain, Duct Leakage Losses), the supply air temperature to the air conditioned space is assumed to equal the condition of the air leaving the apparatus.

The supply air temperature (t_{sa}) can be obtained as follows:

Graphical Method (Use of Psychrometric Chart)

- i. Plot indoor design conditions (db°F & %rh), outdoor design conditions (db°F & wb°F) and connect the two points by straightedge. The line drawn between the two points is proportional to the Mixed Air entering the apparatus.
- ii. Assume the following percent Outdoor Air to Total Supply Air.

% Outdoor Air to Total Supply Air	Facility
8 to 10	Offices
15	Prayer Halls, Restaurants
10 to 15	Substation Control Room

- iii. The line between the indoor design conditions and mixture conditions is proportional to the % Outdoor Air to the Mixed Air. If 10% outdoor air is used, the mixture point is located by measuring one tenth of the line (connecting the indoor and outdoor conditions) from the indoor design conditions.
- iv. Connect Alignment Circle and RSHF line in the Chart.
- v. Draw the actual RSHF line through the room design conditions parallel to the line obtained in Step (d).
- vi. Connect Alignment Circle and GSHF Line in the Chart.

- vii. Draw the GSHF line through the mixture condition up to saturation parallel to the line obtained in Step (f).
- viii. The intersection of the GSHF line and the saturation is the apparatus dew point.
- ix. The intersection of the GSHF line and the RSHF line is the temperature of the supply air to the space and also the temperature of the air leaving the apparatus.
- x. Use this temperature and calculate for the tentative air quantity required to satisfy the room sensible load using the following equation.

$$CFM_{sa} = \frac{\text{Room Sensible Heat (Item 36)}}{1.1 (t_{rm} - t_{sa})}$$

where:

CFM_{sa} supply air design flow
 t_{rm} room air dry bulb temperature, °F
 t_{sa} supply air dry bulb temperature, °F

- xi. Solve also for the CFM_{sa} required to satisfy the Grand Total Heat (Total Equipment Load) by the following equation:

$$CFM_{sa} = \frac{\text{Grand Total Heat (Item 44)}}{4.5 (h_m - h_{la})}$$

Where

h_m mixture of outdoor and return air Enthalpy
 or apparatus entering air enthalpy, Btu/lb
 h_{la} apparatus leaving air enthalpy, Btu/lb

- xii. The values of CFM's obtained in the above two equations shall be verified balance. If not, assume another percentage of outdoor air and repeat the same process until the CFM's are balanced.

HVAC Load Calculation Summary

[illegible]

4.2.2 Heating Load

The purpose of heating load calculations is to estimate the heat loss of the conditioned space at the winter outside and inside design conditions.

Heat loss can be classified into two groups; external heat loss through the building envelope, and the heat required to raise the temperature of infiltration and ventilation air up to the level of conditioned space temperature.

The heat load calculations are described in Cooling and Heating Load Estimate Sheet National Grid Saudi Arabia Form No. 16305/07.

5.0 SYSTEM SELECTION

5.1 General

The evaluation and selection of an air conditioning system to meet a specific need is essential task that must be performed by the qualified and competent Designer. All required factors must be analyzed, judged, screened and coordinated. The foremost considerations are the desires of the owner, occupant and the economic aspects.

5.2 Data Collection

In preparation for the collaborative evaluation, the Designer must list objective, historic and subjective concerns.

5.2.1 Objective

- First Cost
- Capacity
- Space Requirement
- Occupancy
- Annual Energy Use

5.2.2 Historic

The accumulation of data of this type is often difficult to extrapolate, because each system installation (as opposed to equipment installation) may differ from one to the other.

- Maintenance expense
- Component Life
- Expected Downtime

5.2.3 Subjective

- Appearance
- Maintainability
- System Reliability
- Flexibility
- Total Comfort Results

5.3 HVAC System Evaluation

In the HVAC System Evaluation Table (Table 3.0) the parameters to be considered in the evaluation of possible system alternatives are listed. These parameters include dissimilar items, such as costs and operating characteristics, therefore, these parameters are not readily measurable on a common scale. However, if each of the cost items is assigned a Performance Factor (PF) to establish the Relative Performance Points (RPP) of the item, the alternative systems with their various parameters can be evaluated objectively.

Within such a performance factoring system 100 Relative Performance Points (RPP) might be assigned to each parameter initially, then the assigned points can be decreased according to the relative performance level of the particular parameter as compared to the performance of that parameter in other systems being considered. In addition to bringing the parameters to a common scale the relative importance of each parameter also has to be considered.

Establishing the Relative Importance Factors (RIF) could be the most critical phase of the evaluation process, therefore, it requires proper consideration and close coordination with the owner/operator of the system.

The total annual cost in Table 3.0 has to be calculated according to the Present Worth economic comparison method. This method compares the annual cost of the different options, expressed as cash requirement in terms of today's money value. To calculate the present worth of annual costs, the cost has to be multiplied by the Present Worth Factor (PWF). PWF's are available from amortization tables or they can be calculated according to the following equation:

$$PWF = \frac{[(1+i)/(1+j)]^n - 1}{1 - [(1+i)/(1+j)]}$$

Where:

j	=	annual escalation rate of cost item, decimal
i	=	annual interest rate on owner's alternative investment, decimal
n	=	expected useful life of system, usually 20 years.

When the interest rate and escalation rate of the cost are equal, the PWF equals the expected useful life period (n).

5.4 Types of Cooling Systems

The air conditioning systems commonly used by National Grid Saudi Arabia are:

5.4.1 Direct Expansion System

- a. Multiple Packaged
- b. Outdoor Unitary Equipment (Roof-Top)
- c. Indoor Unitary Equipment
- d. Split Type
- e. Window Air Conditioner
- f. Through-the-Wall Air Conditioner

5.4.2 Chilled Water System

The selection of a cooling source requires an analysis of the various alternatives. If chilled water from an existing central plant is available in sufficient quantities and in close proximity, it shall be considered the first choice for a cooling source. The second choice shall be dedicated refrigeration equipment.

Design parameters for central chilled water supply are as follows:

- a. $\Delta t = 10^{\circ}\text{F}$ where entering water temperature (EWT) at 12.22°C (54°F) and leaving water temperature (LWT) at 6.67°C (44°F)
- b. Maximum pressure drop through system 138.6 kPa (20 psi)
- c. Water flow rate at 0.043 lps/kW (0.151 lps/ton) (2.4 gpm/ton)

If the above design parameters are not suitable for a specific application, the responsible utility organization should be contacted for more detailed information on a specific location.

Dedicated refrigeration equipment shall use air cooled condensers, except where an analysis of a project specific conditions, or site utilities, or cooling load characteristics justify condensing water (sea water or cooling towers). Use of water cooled condensers requires an evaluation of the fouling characteristics of the available water.

In remote locations, the use of water chillers is restricted to locations where adequate water supply for system make-up is available.

TABLE 3.0 - HVAC SYSTEM EVALUATION

	RIF	SR	PF	RPP	RIFxRPP	SR	PF	RPP	RIFxRPP
Initial Cost									
P.W. of Annual Cost									
Operating Feature #1		X	X			X	X		
Operating Feature #2		X	X			X	X		
Operating Feature #3		X	X			X	X		
T O T A L	100	X	X	X		X	X	X	
RANKING OF									

6.0 EQUIPMENT SELECTION

This section prescribes guidelines in the selection of HVAC equipment and other system components.

6.1 Direct Expansion (DX) and Chilled Water Coil

- 6.1.1 DX coil (Evaporator) distributor nozzle, expansion valve and chilled water coil selection procedures to be used shall be per manufacturer's methods.
- 6.1.2 The cooling coil shall be selected to produce a desired effect on the air passed through it, in accordance with the sensible, latent and total cooling loads calculated for the space and with the condition of the air entering the coil. However, the final selection defines also the required chilled water flow, the pressure drop at that flow and the required entering water temperature; or in the case of direct expansion coil, the refrigerant temperature. Air side and refrigerant side or chilled water performance side should be considered separately and then matched to produce the final economically optimum coil selection.
- 6.1.3 Fin spacing shall not be closer than 5.51fins/cm. (14 fins/ inch) for evaporator coil and 6.3 fins/cm (16 fins/ inch) for condenser coil.

- 6.1.4 Evaporator and condenser coils shall be copper tubes with aluminum fins except at offshore and near shore locations. Copper fins on copper tubes shall be used for evaporator coils within one km. of sea and for all condenser coils within 1/2 km of sea.
- 6.1.5 Coil return bends shall have flux removed and be protected from corrosion.
- 6.1.6 The evaporator or cooling coil capacity shall not be oversized unnecessarily.
- 6.1.7 Packaged central-station air handling units shall be factory-encased assembly. It shall consist of filters, cooling/dehumidifying coil, fan and the following as required:
- Mixing box
 - Humidifier
 - Electric heater
 - Face damper
 - By-pass damper
- 6.1.8 The velocity of liquid in water coils shall be between 0.7 and 2 mps (2.29 and 6.56 fps). Water pressure drop through coils, water chillers or condensers shall not exceed 100 kPa (35 ft. wg. 14.5 psi). This value applies to chillers, condensers, and cooling coils at their rated fouling factor, unless a lower pressure drop is specified in the design documents. Turbulators shall not be used due to cleaning requirements.
- 6.1.9 The air velocity through coils shall not exceed 2.54 m/s (500 fpm), or a wet coil pressure drop greater than 38 Pa (0.15 in. wg.) per row of coil. Dry coil pressure drop shall not exceed 25 Pa (0.10 in. wg.) per row of coil.

6.2 Mechanical Refrigeration

6.2.1 General

- Mechanical Refrigeration equipment shall conform to the requirements of Uniform Mechanical Code latest issue.
- Refrigeration applications up to 60 tons generally utilize either reciprocating or scroll compressors. Above 60 tons to 200 tons either reciprocating compressors or other types such as rotary helical/screw or centrifugal water chiller machines are used. Above 200 tons rotary helical/screw, centrifugal water chillers or absorption machines are normally used.
- The main refrigerants to be used in air conditioning compressors are non-ozone-depleting refrigerants.

- d. The classification of an air conditioning refrigeration machine is:
 - i. A compressor unit consisting of a compressor, motor, and safety controls mounted as a unit.
 - ii. A condensing unit consisting of a compressor unit plus an inter connected water cooled or air cooled condenser mounted as a unit.
 - iii. A water chilling unit consisting of either a compressor unit or a condensing unit, plus an interconnected water chiller and operating controls mounted as a unit.
- e. The three major types of refrigeration compressors used in large systems are:
 - i. Reciprocating / Scroll
 - ii. Rotary Helical/ Screw
 - iii. Centrifugal

6.2.2 Reciprocating Compressor

Most reciprocating compressors are single acting, using trunk type pistons driven directly through a pin and connecting rod from the crankshaft. The types of reciprocating compressors are:

- a. Open Type - are those in which the shaft extends through a seal in the crankcase for an external drive.
- b. Hermetic - are those in which the motor and compressor are contained within the same pressure vessel, with the motor shaft integral with the compressor crankshaft and the motor in contact with the refrigerant.
- c. Semi-Hermetic - are those of bolted construction (bolted, accessible, amenable to field repair or serviceable).
- d. Welded Shell (Sealed) - are those in which the motor compressor is mounted inside a steel shell which in turn is sealed by welding.

Hermetic and semi-hermetic compressors are often preferred.

6.2.3 Helical Rotary/Screw Compressor:

- a. Screw compressor shall be semi-hermetic, direct drive, positive displacement compressor.
- b. The machine shall consist of two mating helically grooved rotors, a male (lobes) and female (gullies), in a stationary housing with suitable inlet and outlet gas ports.
- c. Compression is obtained by direct volume reduction with pure rotary motion.
- d. Compressor motor shall be constant speed, 3600 RPM, hermetically sealed, suction gas cooled with solid state sensor and electronic winding overheating protection.
- e. Compressor shall have automatic capacity reduction equipment consisting of capacity control slide valve gas discharge pressure and solenoid valve. Capacity modulation shall be continuous from 100% to less than 10%. Compressor shall start unloaded.
- f. The oil separator is mounted integral with the motor housing and act as the final stage oil separator.

6.2.4. The factors affecting the compressor unit selection are:

- Capacity
- Evaporator Temperature
- Condensing Temperature
- Refrigerants
- Sub cooling of the Condensed Refrigerant
- Superheating of the Suction Gas
- Refrigerant Line Pressure Drops
- Operating Limits
- Heat Rejection

6.2.5 Air- Cooled Condensing Unit

- a. Manufacturer's catalogues selection procedure and performance data shall be consulted in the selection of the unit.
- b. The unit must be located so that the flow of air to and from the condenser coil is not impeded. There must be enough space surrounding the unit to prevent recirculation of air.
- c. When selecting condensing units, consideration should be given to the variation in the load and the types and steps of capacity control available on the unit.

- d. The condenser shall be selected to match the cooling coil or evaporator. The condenser capacity shall never be less than the cooling coil or evaporator capacity.
- e. The balance capacity of condenser and evaporator shall match or exceed the space cooling load or requirements.
- f. Air Cooled Condensing Unit is preferred over Water Cooled Condensing Unit.
- g. Packaged, air cooled, reciprocating condensing units shall require only refrigerant piping and electrical connections in order to be placed in operation.
- h. Each condensing unit shall be complete with all operational equipment, including solenoid valve, sight glass, and cartridge type filter on liquid lines. It shall be mounted on a base frame.

6.2.6 Water Chilling Unit

- a. The factors influencing the selection of a water chilling unit are; Capacity, Chilled Water Quantity, Temperature Difference.
 - i. Capacity in kW = $4.19 \times \text{lps} \times \Delta t$ (Btu/Hr = $500 \times \text{gpm} \times \Delta t$)
 Tons of Refrigerant = $\frac{\text{Capacity in kW}}{3.516}$ $\frac{(\text{Capacity in Btu/hr})}{12000}$

 Where: Δt - Temperature Difference between the water temperature entering and leaving the chiller, °C (or °F)

 lps - design water flow, liter per second

 gpm - design water flow in gallon per minute
 - ii. Water Temperature Difference
 - iii. Fouling Factor: Shall be no less than $0.000088 \text{ m}^2\text{-}^\circ\text{C/watt}$ ($0.00050 \text{ ft}^2\text{-hr-}^\circ\text{F/Btu}$) in the cooler with air entering condenser at ambient temperature of the locality. Note that as fouling factor is increased, both chiller capacity and compressor power decrease.
- b. Manufacturer's catalogue, selection procedure and performance data shall be consulted in the selection of chilling units.

6.3 Fans (Air Handling Units)

6.3.1 Types

- a. Centrifugal - used for most comfort air conditioning applications because of its wide range of quiet, efficient operation at comparatively high pressures.
 - Forward Curved Blade
 - Backward Curved Blade (Airfoil and Backward Inclined)
 - Radial (straight)
- b. Axial - used for large air volume applications where higher noise levels are of secondary concern. Often used for industrial air conditioning and ventilation.
 - Propeller (Disc)
 - Tube axial
 - Vane axial
- c. The airfoil blade fan is a high efficiency fan because its aerodynamically shaped blades permit smoother flow through the wheel. It is normally used for high capacity, high pressure applications where power savings may outweigh its higher first cost.

6.3.2 Classes of Construction (Centrifugal Fans)

Table 4.0-Classes of Construction (Centrifugal Fans)

FAN CLASS	MAXIMUM STATIC PRESSURE
A	0 - 750 Pa (0 - 3 in. wg.)
B	750 - 1375 Pa (3 in. wg. - 5.5 in. wg.)
C	Above 1375 Pa (Above 5.5 in. wg.)

6.3.3 Selection

- a. The selection of a fan for a particular air distribution system requires the fan pressure characteristics to fit the system pressure characteristics.

- b. After a decision has been made on the proper type of fan for a given application, the best size selection of that fan must be based on efficiency since the most efficient operating range for a specific line of fans will normally be the quietest. Low outlet velocity does not necessarily assure quiet operation.
- c. Fans with steep pressure-volume characteristic shall be selected for constant air volume distribution system or system that requires an air volume relatively independent of changes in system resistance.
- d. Fans with comparatively flat pressure-volume characteristics are appropriate for variable air volume (VAV) distribution system.
- e. Fan manufacturer's catalogues, performance data in either the graphic curves form or the tabular form shall be consulted in the sizing and selection of fans.
- f. In VAV system, inlet vane damper shall be provided at the fan inlet, or variable speed motor shall be used.
- g. Variable pitch (adjustable) sheaves shall be specified for fan motors for fan speed adjustments to meet actual project conditions.
- h. A minimum of 10% additional air capacity above the total design quantity of air supplied by diffusers and registers in the conditioned spaces shall be allowed for air losses through ductwork, equipment housings, connections, etc.
- i. Access holes for tachometer readings shall be provided in all belt guards. Taking tachometer readings shall be made possible without removal of the belt guard.

7.0 AIR DISTRIBUTION SYSTEM

The first item a Design Engineer of an air distribution system has to consider is the type of system most suitable for the particular application at hand. Another item to be covered at the early stage of the system's design is the air diffusion system of the conditioned space including selection of air outlet types, locating and specifying them. Only after the above items have been finalized, the designer can proceed with the detailed design of the air distribution system. A proper selection process consists of a carefully prepared and well-presented evaluation of the different options available

7.1 System Configuration

Selecting type of air distribution system for a specific project may be one of the most difficult tasks of the HVAC Design Engineer. A proper selection process consists of a carefully prepared and well presented evaluation of the different options available. In this evaluation the economic considerations such as first cost and annual operating cost are the most important aspects, although advantages of certain operating characteristics such as better temperature or humidity control may outweigh the economic disadvantages of a particular system. This is why weighing the operating characteristics of different systems according to the user's preference is so important.

7.2 Air Distribution Systems

7.2.1 Constant Volume, Single Stream Systems

In its simplest and most common form the constant volume, single stream system supplies a constant volume of air to a single zone of conditioned space. Capacity control of this type of system is achieved by varying the amount of heat removal from (or addition to) the air stream.

7.2.2 Dual Stream Systems

To cater for the varying needs of the different zones in large buildings the dual stream systems are frequently used. Two variations of this type of system are the multi zone and double duct systems.

7.2.3 Variable Air Volume Systems

As opposed to the dual stream systems, the variable air- volume systems operate on the principal of constant supply air temperature at varying supply air flow rate. This concept may be put into practice by varying the individual supply air streams to the different zones, by changing the total supply air flow rate to the whole system or a combination of the two methods.

The main problem associated with these systems is the difficulty of providing acceptable air diffusion patterns in the conditioned space with the varying air supply. Other deficiencies of these systems include the inability of providing simultaneous cooling and heating to the different zones of the conditioned space and the lack of close dehumidification control.

7.2.4 Hybrid System

Most of the modern, more sophisticated air conditioning systems combine a form of the variable air volume concept with one or more features of the other system types.

One of these is the variable air-volume reheat system. In this system the total supply air rate is calculated on the basis of maximum instantaneous peak load of the total system rather than the sum of each zone peak loads. Temperature control is achieved in two steps; supply air volume reduction as a first step followed by reheat after the minimum supply air rate is reached.

7.3 Space Air Diffusion

Proper diffusion of air in the conditioned space is an essential feature of a well-designed HVAC system. Conditioned air arrives to the air outlets at a higher velocity and at a higher or lower temperature than it is acceptable in the occupied areas of the space.

7.3.1 Air Outlet Types

Air outlets are most often classified according to their construction features as grilles, slot diffusers, ceiling diffusers and perforated ceiling panels. These four types of outlets widely vary in their primary air diffusion and secondary air induction characteristics.

7.3.2 Location of Air Outlets

According to their location, air outlets are classified into three general categories; sidewall, ceiling and floor mounted outlets. During a performance test described in ASHRAE Fundamentals, air outlets were divided into five groups according to their location and discharge patterns.

- a. Group A outlets include high sidewall grilles, sidewall diffusers and ceiling diffusers with horizontal air discharge. This group of air outlets is recommended for cooling and light heating application only. They are particularly suitable for high airflow rates and large temperature differentials.

- b. Group B outlets are mounted in or near the floor with vertical, non-spreading discharge jet. When the mixture of primary and room air strikes the ceiling, it spreads under the ceiling in all directions before dropping down into the occupied zone.
- c. Group C outlets are also mounted in or near the floor similarly to group B outlets, but they have wide spreading jets and good diffusion capabilities. These outlets basically behave similarly to group B outlets with the difference of larger stagnant air region during cooling, but a better temperature equalization during heating. These outlets are the most suitable for moderate cooling and severe heating application.
- d. Group D outlets include floor or low sidewall mounted air outlets with horizontal air discharge. These types of outlets are mostly used for process installation and they are not recommended for comfort application.
- e. Group E outlets are ceiling diffusers and high sidewall mounted outlets with vertical downward air discharge.

7.3.3 Design of Air Diffusion System

The recommended procedure of an air diffusion system design consists of the following:

- a. Determination of room dimensions and supply air volume from architectural drawings and cooling load calculations.
- b. Selection of the tentative air outlet type and location in the room according to the desired architectural appeal and location requirement of the particular application.
- c. Determination of the room's characteristics length (L). The rooms' characteristic length is the length of the room in the direction of the airflow or the distance to mid plane between outlets, or a variation of the above, depending on the type and location of outlets. Characteristic room lengths for several diffuser types are listed in ASHRAE Fundamentals.

- d. Selection of T_v/L ratio. T_v is the throw distance of an air jet at " v " velocity. For describing comfort criteria of an occupied space the Air Diffusion Performance Index (ADPI) has been developed. ADPI indicates the percentage of locations in the occupied zone where the effective draft temperature and air velocity values are at an optimum. The effective draft temperature is a combination of local temperature differences from the room average temperature and found to be an optimum between -1.7°C (-3°F) and $+1^{\circ}\text{C}$ ($+2^{\circ}\text{F}$). The accompanied optimum air velocity is below 21 mps (70 fps).

ASHRAE Fundamentals summarizes the recommended T_v/L values at 15 mps (50 fps) air velocity and at maximum and above minimum ADPI values for various air outlet devices.

- e. Calculation of throw distance (T_v). T_v can be calculated by multiplying the recommended T_v/L ratio and the room's characteristic length (L).
- f. Selection of the air outlet size from manufacturer's catalog for the above calculated T_v . Other performance criteria, such as noise and pressure drop shall also be satisfied by the selected air outlet device.

AIR DIFFUSION SYSTEM DESIGN DATA

	ROOM NUMBER			
Room Length x width, m				
Room Area, m^2				
Char. Length (L), m				
Supply Air To Room, lps.				
Number of outlets				
Supply Air To Outlet, lps				
Outlet Size				
Room Cooling Load, kW/m^2				
T_v / L				
Calculated T_v				
Actual T_v				
Sound Level, dB				
Pressure Drop, Pa				

7.4 Duct Design

7.4.1 Design Methods

The three duct design methods commonly used for comfort air conditioning systems are the equal friction loss, static regain and velocity reduction methods. The fourth, constant velocity duct design method is used for particulate exhaust systems, and it is not discussed here.

With the equal friction loss method, ducts of the air distribution systems are sized for a uniform pressure drop per unit length of duct. The choice of friction constant depends on the size of air distribution system and noise considerations. Suggested velocity and friction rate design limits are given in ASHRAE Fundamentals. This system is most applicable where the lengths of branch runs are symmetrical with nearly uniform lengths. In systems with unequal branches, headers and branches of equal lengths maybe sized with the equal friction loss methods and the branches with unequal lengths can be sized according to the available pressure at take-offs.

The static regain method is based on maintaining a more or less constant static pressure throughout the air distribution system. At each take-off the increase in static pressure (static regain) is calculated to offset all or part of the pressure drops in succeeding sections. The principal advantage of this method is the relatively constant static pressure at each air outlet in the system, therefore the same outlet devices can be used for equal air outlet rates. The static regain method is more time consuming than the equal friction loss method, but in larger systems the benefit of increased effectiveness outweighs this disadvantage.

When the velocity reduction method is used, the fan discharge duct is sized with the highest system velocity and the successive duct sections are sized with progressively lower velocities. This method should only be used for establishing preliminary duct sizes, and the detailed design of ducting should be performed by the equal friction or static regain method.

7.4.2 Fan Selection

The operating point of the air distribution system's fan is established by superimposing the system curve on fan performance curve.

However, it should be recognized that the actual operating point seldom will coincide with the theoretical intersection point of system curve and fan curve. The discrepancy in terms of airflow may be caused by errors in pressure drop calculation or improper fan inlet/outlet connections.

In addition, the published fan performance curves may not always accurately coincide with the actual fan performance because the fan will not be running in the same system configuration in which it was tested in a controlled condition.

The foregoing defect of the installed system can be rectified with the following steps:

- a. The actual total air flow of the system has to be measured and the actual system curve has to be calculated on the basis of actual operating point established by plotting the air flow measurement on the fan curve.
- b. The fan speed has to be modified for establishing the required operating point at the intersection of the desired air flow and the actual system curve. This can be accomplished by the application of fan laws mentioned below.

The new operating point of the system is established at the intersection of the new fan curve and revised system curve.

7.4.3 Fan Performance

Fan performance at various speed and when handling air can be predicted when the performance is known at a specific condition. The different conditions can be calculated by using the Fan Laws.

- a. Fan Law #1 - Airflow varies in direct proportion to the fan speed i.e. when you increase the speed of a fan by 10%, the airflow also increased by 10%.

$$Q_2/Q_1 = N_2/N_1$$

where,

Q_2	new or required airflow, lps (or CFM)
Q_1	actual or existing airflow, lps (or CFM)
N_2	new or required fan speed, Rad/sec (or RPM)
N_1	actual or existing fan speed, Rad/sec (or RPM)

- b. Fan Law #2 - Static pressure varies by the square of the fan speed i.e. when you decrease the speed of a fan by 10%, the static pressure will decrease by the square of 10%.

$$SP_2 / SP_1 = (N_2 / N_1)^2$$

Where: SP_2 - new or required static pressure, Pa (or in. wg.)
 SP_1 - actual or existing static pressure, Pa (or in. wg.)

- c. Fan Law #3 - Brake power varies by the cube of the speed.

$$BP_2 / BP_1 = (N_2 / N_1)^3$$

where: BP_2 - new or required brake power, kW (or Bhp)
 BP_1 - actual or existing brake power, kW (or Bhp)
 N_2 - new or required fan speed, Rad/sec (or RPM)
 N_1 - actual or existing fan speed, Rad/sec (or RPM)

- d. Fan Law #4 - Fan volume will not change with a change in density. A fan is a constant volume machine and will handle the same airflow regardless of the system fluid or density. The brake power and static pressure will vary in direct proportion to the density. Handling denser or heavier air will produce more pressure and require more power. Fan performance data is based on standard air which has a density of 1.2 kg/m^3 (0.075 lbs. /ft^3).

$$SP_2 / SP_1 = d_2 / d_1$$

$$BP_2 / BP_1 = d_2 / d_1$$

Where: d_2 - new or required density
 d_1 - actual or existing density

7.4.4 Fan/System Curve Relationship

In a given duct system with a known airflow rate when the position of all dampers are stable, a specific, measurable static pressure resistance to the airflow can be determined or measured.

But if the flow rate is increased, the duct system resistance is increased i.e. if the cfm increases the system static pressure increases. The system resistance is the sum of all pressure losses through filters, coils, dampers and ductwork. The system resistance curve or system curve is a plot of the pressure that is required to move air through the system. For fixed system i.e. no changes in damper setting, etc., system resistance varies as the square of the airflow and be calculated as follows:

$$P_2 / P_1 = (Q_{T2} / Q_{T1})^2$$

Where: P_2 - new or required System Static Pressure, Pa (or in. wg.)
 P_1 - actual or existing System Static Press., Pa (or in. wg.)
 Q_{T2} - new or required System Airflow, lps (or CFM)
 Q_{T1} - actual or existing System Airflow, lps (or CFM)

7.4.5 Duct Design Considerations

- a. Splitter type dampers offer little or no control of air volume in ducts. They should be regarded as air diverters only, with maximum effectiveness when present on duct systems exhibiting low resistance to air flow. Valid application of splitter dampers generally occurs at the ends of branch ducts where need to reduce or restrict air flow is not required.
- b. The application of single blade quadrant volume dampers immediately behind diffusers and grilles may tend to throw air to one side of the outlet, preventing uniform air flow across the outlet face or cones.
- c. Manually operated opposed blade or quadrant type volume dampers shall be installed in each branch duct takeoff after leaving the main duct to control the amount of air into these branch ducts.
- d. Turning vanes shall be installed perpendicular to the entering and leaving to minimize air flow turbulence.
- e. Manual volume dampers shall be provided in the duct drop or takeoffs to diffusers and registers to limit the total air to the face damper of the register or neck damper of the diffuser. Sidewall and diffuser dampers cannot be used for reducing high air volumes without inducing objectionable air noise levels.
- f. Outside air louvers can create objectionable air noise on large systems. Louver blades shall be widely spaced with all edges rounded (or double folded) to prevent generation of high pitched air noises. Intake screens should have openings of at least 13 mm (1/2 in.) squares to prevent clogging while offering ample protection against large entering objects.
- g. Manual dampers shall be located downstream of hot and cold zone dampers on each zone of a double duct system.
- h. Double thickness or extended edge turning vanes shall be utilized in all rectangular elbows of return as well as supply ducting.
- i. Furnish extractors where the main air stream in the main duct is a distance away from the branch duct take-off (when located on the inside radius duct wall following an elbow, etc.). Do not specify extractor at branch or main duct takeoffs to provide volume control. Extractors are principally used to divert air to branch duct.
- j. Manual balancing dampers shall be located at each branch duct inlet of vertical duct chases or main return ducts.

- k. Extension ceiling mounted damper hardware shall be provided wherever possible.
- l. Adequate size access doors shall be installed within working distance for volume dampers, pressure reducing valves, reheat coils, mixing boxes, blenders, constant volume regulators, etc., to permit required adjustments.
- m. Avoid placing a return air opening directly in or adjacent to the return air plenum. Sound lining of the duct opening and plenum will not reduce the transmitted noise to acceptable levels.
- n. A slight space or opening between blades of an opposed blade volume damper will generate a relatively high noise level as the air passes through the openings under system pressure. Damper blades shall be sealed with foam rubber or felt to form an effective seal with the blades in the closed position.
- o. Duct leakage may vary from 15 to 45% depending upon workmanship, type of duct construction and fittings, system design, etc. To minimize duct leakage, all duct seams, casing and plenum connections, etc., shall be sealed or taped, thus generally assuring a maximum of 5% duct seam leakage factor.
- p. Avoid the use of masonry or composition wall vertical air shafts for exhaust systems on multi-story buildings.
- q. Indicate volume damper locations at accessible points and wherever possible, a distance from a duct transition or fitting. Care shall be taken during installation to make certain that sheet metal fasteners (screws) do not protrude into the duct and interfere with damper operation.
- r. Do not use air extractors at branch or main duct take-offs to provide volume control. Extractors are principally effective in diverting air to ducts experiencing air shortages, provided these shortages are not due to a considerably higher branch duct resistance to air flow than the other branch ducts on the system. With the condition of higher branch static, the extractor when positioned in the main duct air stream cannot produce sufficient velocity pressure diverted air to overcome the branch duct resistance. When severe dampering is required to build pressure, etc., the high bypass and leakage factors of the extractor nullify its effectiveness.
- s. Proportion the sizes of the duct split fitting or branches based on the air flow requirements of each resulting duct. If higher or lower duct pressure requirements in the branches and outlets are present, adjustment of the proportion of the split shall be made.

- t. To minimize generated duct noise at volume dampers, indicate damper locations at least two diameters from a fitting and as far as possible from an outlet.

7.4.6 Terminal Devices

- a. Avoid locating diffusers or grilles directly into the bottom or side of main air duct. No amount of adjustment will decrease the noise level generated.
- b. Do not design light troffers on the same duct run or zone with standard diffusers or registers due to the greater pressure requirements of troffers which will necessitate excessive throttling at the standard outlets and generate objectionable noise levels.
- c. Restrict use of high induction type diffusers to those applications requiring high air motion.
- d. Avoid long duct runs with large volume diffusers off the main and branch ducts terminating in small diffusers or registers.
- e. Avoid mixing supply registers and diffusers on the same duct section. The greater pressure requirement of the diffusers will necessitate extensive throttling at the registers and generate air noise.
- f. Avoid placing diffusers or registers too close the ceiling or in a manner that air patterns will be discharged into ceiling mounted light fixtures.
- g. If possible, provide adjustable extractors at each duct takeoff to a register.
- h. Return air registers shall be located on or near exterior walls, preferably at or near floor level.
- i. Do not use an outlet with low induction characteristics where the air volume being distributed is high and the distance of throw is short.
- j. Return air grilles shall be selected for operation at low face velocities (0.50 to 3.05 mps or 100 to 600 FPM) to minimize noise levels.
- k. When designing duct drops to diffusers, provide a minimum length of 2 times the duct diameter (or square dimension) in length to assure even distribution from the outlet.
- l. Return air grilles and duct connections which open into common return plenum without air fans shall be oversized when possible.

- m. Avoid the passage of return air from one space or zone through that of another to reach a return air grille.
- n. Avoid the use of built-in door louvers for passage of return air when the supply air system operates at low pressure (ceiling plenum supply, etc.).
- o. Avoid the use of combination supply-return outlets. Air quantities handled by supply and return section of outlet can be accurately measured and adjusted by means of specific balancing procedures. However, percent of short circuiting of supply air to return cannot be determined.

7.5 General Duct Design Procedures

7.5.1 Study the plans of the building and arrange the supply/ return outlets to provide proper distribution of air within each space.

7.5.2 Determine the cooling or heating load to be supplied by each outlet. Adjust calculated actual air quantities for duct heat gains or losses and duct leakage. Also, adjust the supply, return, and/or exhaust air quantities to meet space pressurization requirements.

$$\text{Supply Outlet Flow Rate} = \frac{\text{Outlet Heat Gain}}{\text{Total Heat Gain}} \times \text{Total System Flow Rate}$$

7.5.3 Select outlet sizes from manufacturer's data

7.5.4 Determine ductwork location.

7.5.5 Determine cooling or heating unit location.

7.5.6 Sketch the duct system, connecting the supply outlets and return intakes with the central station apparatus, taking cognizance of the building construction, and avoiding all structural obstructions and equipment. The space allocated for the supply and return ducts often dictates the system.

7.5.7 Make a single line diagram of the supply and return duct systems. Label all fittings and transitions to show equivalent lengths on the single line drawing. Show measured lengths of ductwork on the single line drawing. Determine the total effective length of each branch supply. (Begin at the air handler and add all equivalent lengths and measured lengths to each outlet: Effective Length = Equivalent length + Measured Length).

7.5.8 Label the supply outlet flow rate requirement on the single line drawing for each outlet.

- 7.5.9 Determine the sizes of all main and branch ducts by the selected design method.
- 7.5.10 Calculate the total pressure requirements of all duct sections, both supply and return, and then plot the total pressure grade line. The total pressure shall include losses through air handling apparatus, filters, duct heaters, etc.
- 7.5.11 Layout the system in detail. If significant duct routing and fitting variations have occurred from the original design, recalculate the pressure loss.
- 7.5.12 Analyze the design for objectionable noise levels and specify sound attenuators as necessary.
- 7.5.13 Select the fan.

7.6 Velocity and Friction Rate Design Limits

7.6.1 See ASHRAE Fundamentals.

7.6.2 The following may be used also as a preliminary guide:

**Table - 5.0 : RECOMMENDED DUCT VELOCITIES FOR LOW VELOCITY SYSTEMS, mps
(FPM)**

APPLICATION	MAIN DUCTS		BRANCH DUCTS	
	SUPPLY	RETURN	SUPPLY	RETURN
Residences	3.56 - 4.57 (700 - 900)	3 (600)	3 (600)	3 (600)
Apartments, Hospital Bedrooms	7.62 (1500)	6.6 (1300)	6.1 (1200)	5 (1000)
Private Offices, Director's Room, Libraries	6.1 - 8.1 (1200 - 1600)	4.57 - 6.1 (900 - 1200)	4 - 6.6 (800 - 1300)	3 - 4.57 (600 - 900)
Theaters, Auditoriums	5 - 6.6 (100 - 1300)	4.57 - 6.1 (900 - 1200)	3 - 4.57 (600 - 900)	2.54 - 3.56 (500 - 700)
Dining Halls, Cafeterias	10 (2000)	7.62 (1500)	8.13 (1600)	6.1 (1200)
Industrial Buildings	6.1 - 9.1 (1200 - 1800)	4.57 - 7.1 (900 - 1400)	4 - 5 (800 - 1000)	3 - 4 (600 - 800)

7.7 Typical Design Velocities for Duct Components (See ASHRAE Fundamentals)

<u>DUCT ELEMENT</u>		<u>ACTUAL FACE VELOCITY</u> <u>mps (FPM)</u>
LOUVERS		
A.	Intake	
1.	3.31 m ³ /s (7000 CFM) and greater	2.03 (400)
2.	Less than 3.31 m ³ /s (7000 CFM)	See ASHRAE Fundamentals
B.	Exhaust	
1.	2.36 m ³ /s (5000 CFM) and greater	2.54 (500)
2.	Less than 2.36 m ³ /s (5000 CFM)	See ASHRAE Fundamentals
FILTERS		
A.	Panel Filters	
1.	Viscous Impingement	1.02 - 4.06 (200 - 800)
2.	Dry-Type, Extended Surface	
a.	Flat (Low Efficiency)	Duct Velocity up to 3.81 (750)
b.	Pleated Media (Intermediate Efficiency)	-do-
c.	HEPA	1.27 (250)
B.	Renewable Media Filters	
1.	Moving-Curtain Viscous Impingement	2.54 (500)
2.	Moving-Curtain Dry-Media	1.02 (200)
C.	Electronic Air Cleaners	
1.	Ionizing Plate-Type	1.52 - 2.54 (300 - 500)
2.	Charged-Media Non-ionizing	1.27 (250)
3.	Charged-Media Ionizing	-
HEATING COILS		
A.	Steam and Hot Water	2.54 - 3.05 (500 - 600) 1.02 (200) Min./7.61 (1500) Max.
B.	Electric	
1.	Open Wire	Refer to Mfg. Data
2.	Finned Tubular	Refer to Mfg. Data

<u>DUCT ELEMENT</u>		<u>ACTUAL FACE VELOCITY</u>
		<u>mps (FPM)</u>
DE-HUMIDIFYING COILS		2.54 - 3.05 (500 - 600)
A.	Without Eliminators	3.55(700) and greater
B.	With Eliminators	
AIR WASHERS		
A.	Spray-Type	1.52 - 3.05 (300 - 600)
B.	Cell-Type	Refer to Mfg. Data
C.	High Velocity, Spray-Type	6.09 - 9.14 (1200 - 1800)

7.8 The following is the recommended Outlet (Terminal) Velocity which can be used as a guide in selection of terminal devices.

Table 6.0 - Recommended Outlet Velocity

Application	Terminal Velocity, mps (FPM)
Residences	2.54 - 3.81 (500 - 750)
Apartments	2.54 - 3.81 (500 - 750)
Mosques	2.54 - 3.81 (500 - 750)
Private Offices, Acoustically Treated	2.54 - 3.81 (500 - 750)
Private Offices, Not Treated	2.54 - 4.07 (500 - 800)
Theaters, Auditoriums	5.08 (1000)
General Offices	5.08 - 6.35 (1000 - 1250)

8.0 DUCTWORK

8.1 General Requirements

- 8.1.1 Ductwork shall be designed, fabricated and installed in accordance with latest accepted good practices for sheet metal work and in accordance with SMACNA, NFPA 90A, Specifications and the Drawings.
- 8.1.2 Unless otherwise specified, duct material shall be coated galvanized steel of lock-forming grade conforming to ASTM A-653.
- 8.1.3 All ducts for substations shall be constructed and fabricated considering the following standard:
 - a. Internal/indoor duct shall be factory insulated with 25 mm (1 in.) fiberglass insulation with density of not less than 24kg/m^3 and shall be of the double skinned with slip-on-flanged transversal joints.
 - b. External/outdoor duct shall be factory insulated with 50 mm (2 in.) fiberglass insulation with density of not less than 48kg/m^3 and shall be of the double skinned with slip-on-flanged transversal joints. Ducts shall be epoxy painted for protection against weather conditions and against rust. The duct work shall be covered with 0.8 millimeter thick aluminum sheet cladding.

8.2 Duct Pressure Classification

- 8.2.1 Unless specified or allowed, the minimum thickness for rectangular and round ductwork shall be constructed in accordance with SMACNA HVAC Duct Construction Standard for Metal and Flexible. Duct pressures are classified as shown in Table 7.0.
- 8.2.2 Ductwork layout shall be neat and orderly. If ductworks are exposed maximize head room as much as possible.
- 8.2.3 Turning vanes in all rectilinear elbows shall be single vane construction as recommended by SMACNA. Angle of turning vanes shall correspond to angle of elbow.

Table 7.0 - Duct Pressure Classification Table

	Duct Class	Static Pressure Rating, Pa (in. wg.)	Velocity mps (FPM)	PRESSURE
High Pressure Duct	High Pressure	2500 (10)	10 (2000) Up	Positive
	Medium Pressure	1500 (6)	10 (2000) Up	Positive
	Medium Pressure	1000 (4)	10 (2000) Up	Positive
	Medium Pressure	750 (3)	20 (4000) Down	Positive or Negative
Low Pressure Duct	Low Pressure	500 (2)	12.5 (2500) Down	Positive or Negative
	Low Pressure	250 (1)	12.5 (2500) Down	Positive or Negative
	Low Pressure	125 (0.5)	10 (2000) Down	Positive or Negative

8.2.4 Volume dampers shall be provided wherever necessary for complete control of air flow in all supply and return branches, divisions in main supply and return ducts including each individual air supply outlet and outside air ducts. Multi-blade volume dampers in all zone ducts shall be provided at multizone air handling units. Where access to dampers through a suspended ceiling is necessary, the proper location of the access doors shall be coordinated.

- a. Air extractor shall be provided for branch takeoff (rectangular branch or sidewall grille) not controlled by splitter damper.
- b. Bell mouth connector with volume damper shall be provided in round duct branch takeoffs not controlled by splitter damper.

8.2.5 Dampers used to isolate areas protected by Clean Agent Fire Extinguishing System (including but not limited to raised-floor dampers, return air duct dampers, supply air dampers, pressurization control dampers) shall comply with UL-555S, Leakage Class I requirements. When a damper is in close position it shall not leak more than 20.23 lps per m² (4 CFM per ft²) at 25 mm (1 in.) wg, nor more than 40.48 lps per m² (8 CFM per ft²) at 100 mm (4 in.) wg.

8.2.6 Rectangular ducts which terminate with an open end shall have the framed with galvanized steel angles, with 13 mm (1/2 in.) mesh galvanized hardware cloth cover.

8.3 Flexible Connections

- 8.3.1 Provide flexible fabric connections between each piece of air moving equipment and ductwork. At least 25 mm slack shall be allowed in these connections to insure that no vibration is transmitted from fan to ductwork.
- 8.3.2 The fabric shall either be folded in with the metal or attached with the metal collar frames at each end to prevent air leakage.
- 8.3.3 The flexible material for indoor installation shall be neoprene coated glass fabric, and shall be UL listed, non-combustible, fire retardant, waterproof, airtight, resistant to abrasion and damage from flexing.
- 8.3.4 For outdoor installation, the same material specification as in 8.3.3 above shall be used with two coatings of hypalon.
- 8.3.5 Materials shall have a flame spread and smoke developed rating of 25/50 or less as per NFPA 255.

8.4 Acoustical Liner

- 8.4.1 Provide acoustical liner in low pressure duct for building offices as follows:
 - a. 25 mm (1 inch) thick in all supply ducts in Mechanical Room in which unit is located, and in all supply ducts within 9.15 meters (30 feet) of unit (whichever requirement is greater shall govern).
 - b. 25 mm (1 inch) thick in all return ducts within 6.10 meters (20 ft.) of unit.
 - c. 25 mm (1 inch) thick in all rigid ductwork on the discharge side of Variable Air Volume Boxes.
- 8.4.2 Acoustical Material: Density 48 kg per m³ (3 lbs/ft³), "K" value not more than 0.036 w/m-°C or 0.25 Btu-in/hr-ft²-°F at 24°C (75°F) mean temperature difference, neoprene faced. Maximum friction rate shall be not more than 1.3 multiplier for ASHRAE values.
- 8.4.3 Sheet metal must be 50 mm (2 in.) larger in both directions where liner is installed in ductworks.

8.5 Screen

All fresh air intake openings shall be provided with removable screens fabricated of gauge 16 galvanized steel, 13 mm (1/2 in.) wire mesh with a galvanized steel framing.

8.6 Connections to Louvers

Connections to louvers shall be made watertight. A minimum of 90 cm (3 ft) ductwork behind louver shall have watertight soldered joints and shall be sloped to weep holes in bottom of louver. Where possible, lap duct shall be over bottom louver blade to drain any water that may enter.

8.7 Plenums

Plenums shall be made of galvanized steel framing members and galvanized steel sheets, cross-braced and rigidly braced with galvanized angles. Gauges and bracing shall conform to SMACNA recommendations for ductwork of like size. Opening for fans, access doors, etc., shall be framed with galvanized steel angles. Access doors shall be provided with locking latch.

8.8 Smoke/Fire Dampers

8.8.1 Fire dampers shall be provided in ducts penetrating fire rated walls and shall conform to the requirements of NFPA 90A and SMACNA with recommended steel sleeves, fusible links, spring catches and non-corrosive bearings. Dampers shall be UL listed and shall be installed in a manner which will conform to the conditions of UL listing.

8.8.2 In substations, motorized combination of smoke and fire dampers shall be installed in supply and return duct penetrations or openings on all fire rated walls and floor slabs.

8.9 Access Door to Smoke/Fire Dampers

Access doors to fire dampers in paragraphs 8.8.1 and 8.8.2 above shall be provided in the duct for inspection and service to smoke/fire damper and fusible link. Access doors shall be airtight and conform to the recommendations of NFPA and SMACNA.

8.10 Penetration Closures

8.10.1 Closures or airtight seals shall be provided for all openings in floors or walls where ducts penetrate. Provide a minimum of 88 mm (3-1/2 in.) high watertight concrete curbs around all duct openings in concrete floors or roofs.

8.10.2 The materials of seals shall be as follows:

- a. 18 gauge galvanized sheet metal where no fire rating is required, and where no fire dampers are installed.
- b. 10 gauge galvanized steel where fire rating is required or where fire damper is installed.

- c. For non-fire rated walls, where openings are constricted in a way that will render the use of metal closures impractical, lead wool packing shall be used to seal openings in a manner to be permanently air tight and sound attenuating.

8.11 Flexible Ducts

- 8.11.1 Flexible ducts shall be insulated type, with two ply; woven fiberglass inner lining on vinyl coated spring steel wire helix, fiberglass insulation minimum 25 mm (1 in.) thick, and with outer vapor barrier jacket.
- 8.11.2 Flexible duct shall not exceed 1.83 meters (6 ft.) in length, and shall be installed in as straight line as possible. Bends shall not exceed 90 degrees total, with a minimum radius of 2 times inside diameter. Do not allow any kinks in duct. Maximum friction rate shall be 7.5 mm (0.3 in.) wg. per 30.5 m (100 ft.) duct. Duct shall have UL-181 air duct listing. The total pressure drop for any 1.83 m (6 ft.) section installed shall not exceed 2.54 mm (0.1 in.) wg.

9.0 CONTROL AND ZONING

9.1 Controls

- 9.1.1 HVAC systems shall be designed and installed using commercially available equipment and techniques to provide Automatic Temperature Controls (ATC), and when required, humidity, pressure and air quality. Safety and limit automatic controls such as freeze stats, fire stats, flow switches, smoke detectors, refrigeration low-high pressure cutouts, etc. shall be provided for safe operation of the entire HVAC system. The control system shall be powered by electric, electronic, pneumatic and self-contained controls. All thermostats (humidistat, etc.) shall be manually adjustable over the operating range.

One of the most important components in ATC system operation are the control valves (i.e. the solenoid valves and 2-way or 3-way control valves for DX and hydronic systems respectively) that provide accurate control over the fluid flow through the cooling coil. Since solenoid valves formed integral part of the commercially available A/C unit in the DX system, further discussion on valve sizing is no longer practical.

In hydronic system, the properly sized valve to provide good fluid flow control will be one-half the diameter of the surrounding pipe. A correctly sized valve must provide a significant head loss that can be calculated by the following formula:

$$C_v = Q / (\Delta P)^{1/2}$$

Where: C_v - flow coefficient or valve constant (normally provided by Control Valve Manufacturer)

Q - flow rate of the fluid, gpm

ΔP - pressure drop through a wide open valve, psi

Some typical guidelines used to calculate the desired pressure drop are:

1. The pressure drop through the valve should be equal to the pressure loss through the coil;
2. The pressure drop through the valve should be equal to 1/2 the total pressure drop between the supply header and the return header; or
3. The pressure drop through the valve should be equal to one-third the total system pressure loss.

9.1.2 The performance of HVAC system shall be controlled by a thermostat located in the supply air stream at coil discharge or in the return air stream within the conditioned space. Water flow control shall be by throttling valve in the discharge of water coil.

9.1.3 Controls for heating shall be interconnected with air conditioning system with a minimum of 3.3°C (6°F) dead band between heating cut off and cooling start temperatures for all applications where more precise temperature control is not required.

9.1.4 Air conditioning units shall automatically shut down when fire alarm system is activated in accordance with NFPA 90A and NFPA 101.

9.2 Zoning

9.2.1 Buildings over 9.8 m (32 ft.) wide and buildings with perimeter offices shall be divided into interior and exterior zones.

9.2.2 In dividing an area into zones, similarities of exposures, internal loads and occupancy must be considered. The grouping of spaces into zones should be determined by physical size, arrangement of constituent spaces, and uniformity of control requirements. The character of occupancy, whether executive, supervisory or general, may also govern the zoning.

9.2.3 For a successful zone control, the requirement of cooling and heating, both hourly and seasonally, must be consistent throughout the spaces constituting a zone.

9.3 HVAC Control System (For Substation Building)

9.3.1 The Direct Digital Control (D.D.C.) system shall be compact module type, 24V DC shall contain digital and analogue for input and output points as specified below:

1. All alarms and indications (such as pre-filter, bag filter, exhaust fans, etc.) shall be considered as input points in D.D.C. system.
2. Compact module D.D.C. shall be at least 16 points input and 16 points output.
3. D.D.C. shall be ready to communicate with computer system to building B.M.S.
4. D.D.C. compact module shall be at least 20% free points spare.
5. D.D.C. test set shall be provided by the CONTRACTOR for maintenance purposes.

9.3.2 Provide duct smoke detectors complete with response indicator and shall be connected to the fire alarm control panel under separate zone.

9.3.3 Control System shall also include all the necessary pressure and temperature control protective devices as well as control relays, consisting of, but not be limited to the requirements defined in 73-TMSS-01, and in TES-P-119.19.

10.0 VENTILATION AND EXHAUST SYSTEMS

10.1 General

Generally, the introduction of outside (make-up) air into a building serves several purposes:

- a. To force exhaled air out of the building, to remove moisture from indoor air, to dilute carbon dioxide concentration and to remove body generated odors from the conditioned space.
- b. In addition to the above, for most buildings located in Saudi Arabia it is necessary to provide a slight positive internal pressure for keeping the airborne dust out. In substations, control rooms and other dust sensitive facility, it is mandatory to provide 10% of supply air for fresh air make-up.

10.2 Make Up Air

Make up air shall replenish all exhaust fan losses, air lost through doors and leakage through openings, and shall provide adequate fresh air for the occupants. Make up air shall be filtered and conditioned. The minimum amount of outside air shall equal the greater of 5 percent of supply in excess of all exhaust air or 3.8 L/s (8 CFM) per occupant.

10.3 Ventilation requirements

Ventilation requirements for occupants in various conditioned buildings or facilities shall be referred to ASHRAE Fundamentals.

10.3.1 In residential buildings the ventilation requirement is satisfied by infiltration and no separate outside air make-up is required. The minimum rate of infiltration to be used for heating/cooling load calculations shall be 0.5 air change per hour.

10.3.2 Toilet rooms shall be provided with a mechanically operated exhaust system capable of providing a complete change of air every 6 minutes. System shall exhaust directly to the outside, and the point of discharge shall be at least 1.52 m (5 ft.) from any openable window or door, and 1 m (3 ft.) above and 3.05 m (10 ft.) away from any air intake such that .

Conditioned air supplied to toilets shall be exhausted to the outside of building and shall not be re-circulated to the air handling units.

10.4 Occupied Non-Air Conditioned Enclosed Storehouses

10.4.1 Occupied non-air conditioned enclosed storehouses shall have mechanical forced ventilation. Inlet air intakes shall be designed to minimize sand and dust intrusion. The building design shall have roof insulation and as required on the exterior walls.

Maximum "U" Value

0.34 watt/ m² -°K (0.06 Btu/hr-ft²-°F)

0.56 watt/ m² -°K (0.10 Btu/hr-ft²-°F)

10.4.2 All buildings or portions thereof where flammable or toxic materials are used, shall be analyzed by Industrial Hygienist and by the Industrial Security Performance Development Department for the required ventilation rate.

10.5 Hospital Ventilation and Exhaust Systems

Hospital ventilation and exhaust system shall be designed in accordance with Accreditation Manual for Hospitals or National Board of Fire Underwriters' Pamphlet No. 46.

10.6 Ventilation of Battery Rooms

10.6.1 Conventional Vented Lead-Acid Type Battery

The minimum rate of ventilation shall be designed at 12 air change per hour in accordance with TES-P-119.19.

10.6.2 Maintenance Free Sealed Lead-Acid Type Battery

For safety requirement, the ventilation air flow rate shall be designed at least 8 air change per hour as per TES-P-119.21.

10.6.3 Return air shall not be re-circulated to the air conditioning equipment but shall be exhausted to the outside of the building. Return air ducts or grilles shall not be provided in the battery room.

10.6.4 Ductwork shall be provided at points at or near the floor level (for gases and vapor heavier than air) and at points above the battery (for gases and vapor lighter than air). The ductwork shall be protected with acid and corrosion resistant coatings.

10.6.5 Exhaust fan/motor assembly shall be corrosion proof and explosion proof ideal for 24 hours continuous operation. Motor shall be NEMA design Class B insulation.

The table shown below can be used as a guide in determining the recommended air changes for a particular space or room.

Table 8.0 - Recommended Air Change Rates

TYPE OF SPACE	AIR CHANGE PER HOUR *	TYPE OF SPACE	AIR CHANGE PER HOUR*
Assembly Hall	10 - 15	Laundry	15 - 25
Auditorium	4 - 15	Library	3
Battery Room	12	Locker Room	6
Boiler Room	10 - 15	Machine Shop	6
Mosque	10 - 15	Offices	8 - 30
Classroom	10 - 12	Pump Room	8 - 10
Corridor	3 - 12	Recreation Room	8 - 30
Developer	10 - 60	Restaurant	6 - 12
Dining Room	4 - 6	Shop General	6 - 20
Engine Room	4 - 6	Substation Electric	5 - 8
Garage	10 - 15	Toilet	6 - 10
Generating Room	12 - 30	Transformer Room	12
Heat Treating	60 - 99	Turbine Room	10 - 30
Kitchen	10 - 20	Warehouse	2 - 10

NOTE:* The air change rates must conform to the Local Health Department code covering the type of installation

11.0 SOUND AND VIBRATION CONTROL

11.1 Sound Control for Outdoor Equipment Installations

The following factors shall be considered to assure quiet installations of outdoor equipment:

- 11.1.1 The equipment shall be located as far as practical from points of possible complaints. Installation in confined or restricted locations, such as courtyards and alley- ways, shall be avoided.
- 11.1.2 The equipment shall be oriented so that maximum sound radiation patterns are directed away from points of possible complaint. (Maximum sound radiation usually occurs in the direction that air enters and leaves the equipment).
- 11.1.3 Natural and, where necessary, artificial barriers may be used to reduce the sound that reaches points of possible complaints. To be effective, barriers must be of solid, heavy construction and large enough so that flanking effects do not minimize their benefits.
- 11.1.4 Silencers may be used to provide desired sound reduction if the equipment is capable of handling their air flow resistance. It is advisable to obtain the equipment manufacturer's recommendations on their use. The resulting sound levels will be the equipment levels at the new operating conditions minus the net insertion loss provided by silencers.
- 11.1.5 The equipment shall be installed in accordance with the manufacturer's instructions and checked for proper acoustical, as well as mechanical, operation. For continued acceptability, regular maintenance procedures shall cover items that could produce unnecessary sound, such as the tightening of loose parts and the replacement of worn or damaged components.

11.2 Sound Control for Indoor Air Conditioning Systems

The following are recommended procedures for sound control calculations:

- 11.2.1 Choose the appropriate design noise level for each room, or class of rooms, dependent on use and construction.
- 11.2.2 Select equipment radiating into the room that will meet this design level.
- 11.2.3 Calculate the required duct attenuation to reduce central system noise to design levels.
- 11.2.4 Design or select sound attenuating duct elements to meet requirements established in step 3.

- 11.2.5 Repeat steps 2, 3 and 4 for the return side.
- 11.2.6 Design mechanical room, or rooms, to properly attenuate the generated noise from the equipment contained.
- 11.2.7 Prevent the transmission of annoying vibration through the structure to occupied spaces.
- 11.2.8 Design areaways and fresh air intakes to meet outdoor criteria. Also check outside to inside path from this source.

11.3 Mechanical Equipment Room Noise Isolation

- 11.3.1 Locate mechanical room remote from all acoustically critical surfaces. Buffer zones (storage rooms, corridors or less noise-sensitive spaces) may be placed between all mechanical equipment rooms and rooms requiring quiet environment.
- 11.3.2 Construction enclosing the mechanical equipment room shall be poured or masonry units of sufficient surface weight to provide adequate sound transmission loss capability. Walls must be airtight and caulked at the edges to prevent sound leaks.
- 11.3.3 Wherever ducts, pipes, conduits, etc., penetrate the walls, floor, or ceiling of a mechanical room, it is necessary to acoustically treat the openings for adequate noise control. A 12 to 16 mm clear space should be left all around the penetrating element and this space should be filled with fibrous material for the full depth of the penetration. Both sides of the penetration should be sealed airtight with a non-hardening resilient sealant.
- 11.3.4 Doors into mechanical equipment rooms are frequently the weak link in the enclosure. Where noise control is important, they shall be as heavy as possible, gasketed around the perimeter, without grilles or openings and be self-closing.

11.4 Recommended Indoor Design Goals

Recommended Indoor Design Goals for Air Conditioning System Sound Control*
(Note: These are for unoccupied spaces, with all systems operating.)

	<u>Type of Area</u>	<u>Recommended RC OR NC Criteria Range</u>
11.4.1	Offices	
	a. Board room	20 to 25
	b. Executive	25 to 30
	c. Conference rooms	25 to 30
	d. Private offices, reception rooms	30 to 35
	e. General open offices, Drafting rooms	35 to 40
	f. Computer/business machine areas	40 to 45
	g. Halls and corridors	35 to 45
11.4.2.	Hospitals and Clinics	
	a. Private rooms	25 to 30
	b. Wards	30 to 35
	c. Operating rooms	25 to 30
	d. Laboratories	30 to 35
	e. Corridors, halls, lobbies, waiting rooms	30 to 35
	f. Washrooms and toilets	35 to 40
11.4.3.	Masjids (Mosques)	25 to 30**
11.4.4.	Schools	
	a. Lecture and classrooms	25 to 30
	b. Open-plan classrooms	30 to 35**
11.4.5.	Libraries	30 to 35

* Design goals can be increased by 5 dB when dictated by budget constraints or when noise intrusion from other sources represents a limiting condition.

** An acoustical expert shall be consulted for guidance on these critical spaces.

RC: Room Criterion (Sound Pressure level, dB recommended 20 MicroPascals)

NC: Noise Criterion (Sound Pressure level, dB recommended 20 MicroPascals)

11.5 Vibration Isolation and Control

The following general procedure shall be followed to avoid problems of vibration and structure-borne noise transmission:

- 11.5.1 Evaluate the inherent quietness of the various types of equipment and select the types with the lowest sound and vibration levels, consistent with engineering and cost considerations.
- 11.5.2 Locate equipment rooms so they are not directly adjacent to, above, or below areas which are critical from a noise and vibration standpoint. Equipment with inherently large unbalance or vibratory forces shall be installed at grade or remote basement locations whenever possible.
- 11.5.3 Locate pipe and duct shafts in utility or service cores near non-critical areas such as elevator shafts, stair- wells, and toilets, rather than adjoining critical areas such as bedrooms or private offices.
- 11.5.4 Design supporting structures to be as stiff as possible. Although most equipment room floors are usually 10 or more times stiffer than equipment isolators, they are capable of deflections resulting in floor natural frequencies in the operating speed range of most HVAC equipment. To avoid problems when large equipment or equipment capable of generating substantial vibratory forces is to be located near noise-sensitive areas, either perform a dynamic analysis or retain the services of an acoustical consultant.
- 11.5.5 Specify maximum allowable equipment vibration levels.
- 11.5.6 Provide spring vibration isolators for equipment.

11.6 Selection of Vibration Isolators

Vibration isolators shall be free standing, un-housed, laterally stable springs wound from high strength spring steel. Springs shall have a lateral stiffness greater than 0.8 times the rated vertical stiffness and shall be designed to provide up to 50% overload capacity. Springs shall be supported either with a neoprene cup or a metal base plate complete with a ribbed neoprene pad, minimum 6 mm (0.25") thick, bonded to the base plate. Springs shall be selected to provide operating static deflections as required.

12.0 AIR FILTERS

12.1 Air Handling Unit Filter

Air handling unit filters shall be high efficiency bag filter or washable material with a minimum Dust Spot Efficiency of 25%, and 90% weight arrestance rating.

12.2 Make-up (Outdoor) Intake System

Make-up (Outdoor) Intake System shall consist of Sand trap louver, pre-filter (inertial filter), or viscous oil filter, or media-filter capable of 25% min. Dust Spot Efficiency and 90% weight arrestance (ASHRAE).

12.3 Air Velocity Through Pre-Filters

The air velocity through pre-filters shall not exceed 3.05 mps (600 FPM) and dirty pre-filter pressure drop shall not be greater than 125 Pa (0.50 in. wg.).

12.4 Air Velocity Through High Efficiency Filters

The air velocity through high efficiency filter shall not exceed 1.3 m/s (255 fpm) and dirty filter pressure drop shall not be greater than 250 Pa (1.00 in. wg.).

12.5 Selection Guidelines

12.5.1 Have air flow as uniform across the filter face as possible.

12.5.2 Consider the use of prefilter with high-efficiency units for longer service life.

12.5.3 Provide weather louvers with trash screens on intakes.

12.5.4 Use a draft gauge on constant volume systems to determine when a filter should be serviced.

12.5.5 Provide sufficient access for cost-effective servicing.

12.5.6 Select filters carefully when variable air volumes are involved. (Volume less than 20% or more than 30% of normal rating maybe encountered which require careful selection for optimum performance.)

12.5.7 Insist on support data for efficiency and life estimates.

12.5.8 Do not over rate air filter beyond manufacturer's recommendations.

12.5.9 Do not exceed manufacturer's recommended final resistance values for any given filter.

- 12.5.10 Do not forget lint in atmospheric dust is difficult to remove from viscous impingement filters and can cause electrical short circuit in electrostatic air cleaners.
- 12.5.11 Do not forget horsepower cost requirements for high- resistance filters.
- 12.5.12 Do not consider only "first-cost" when selecting an air filter.
- 12.5.13 Do not install HEPA filters without adequate prefilters.

13.0 INSULATIONS

13.1 Ducts

- 13.1.1 All conditioned air supply and return ducts, regardless of length shall be insulated. Supply and return air ducts, exposed to unconditioned space, shall also be insulated. Insulation shall be in accordance with Uniform Mechanical Code (UMC). All insulation materials shall have a mold, humidity and erosion resistant face that meet the requirements of UMC Standard. Materials shall have a flame spread rating of not greater than 25 and a smoke developed rating of not greater than 50 in accordance with NFPA 255. Supply and return air ducts, not acoustically lined, shall be insulated externally.

13.1.2 Ducts (Not Acoustically Lined)

Fiberglass; minimum density 24 kg/m^3 (1.5 lbs/ft^3); thermal conductivity "K" value not more than $0.036 \text{ w/m-}^\circ\text{K}$ ($0.25 \text{ Btu-in/hr-ft}^2\text{-}^\circ\text{F}$) at 24°C (75°F) mean temperature difference; with factory adhered reinforced foil faced flame resistant Kraft paper vapor barrier.

13.1.3 Ducts (Acoustically Lined)

Density: 48 kg/m^3 (3 lbs/ft^3); "K" value not more than $0.036 \text{ w/m-}^\circ\text{K}$ ($0.25 \text{ Btu-in/hr-ft}^2\text{-}^\circ\text{F}$) at 24°C (75°F) mean temperature difference, neoprene faced; maximum friction rate shall not be more than 1.3 multiplier for ASHRAE values.

13.1.4 Flexible Ducts

Two ply; woven fiberglass inner lining on vinyl-coated spring steel wire helix; fiberglass insulation 25 mm (1 inch) minimum thick, with non-combustible polyethylene water vapor barrier jacket.

13.1.5 Ducts (Outdoor)

- a. Inside of duct shall be lined with insulation as per par. 13.1.3. (This is applicable only on areas where critical sound attenuation is required).
- b. Airflow direction shall be marked-up on outdoor ducts for easy identification.

13.2 Refrigerant Lines

13.2.1 Suction Lines

Suction lines shall be insulated to avoid dripping due to high ambient temperatures and simultaneous high relative humidity. However, it is generally desirable to have the suction line capable of absorbing some heat to evaporate any liquid which may have entered the suction line from the evaporator.

13.2.2 Liquid Lines

Liquid lines shall not be insulated if the ambient temperature is lower than or equal to the temperature of the liquid. Insulation is recommended only when the liquid line can pick up a considerable amount of heat.

13.2.3 Hot Gas Lines

Hot gas lines shall not be insulated. Any heat lost by these lines reduces the work to be done by the condenser.

- 13.2.4 Insulation shall be non-combustible closed cell tube; thermal conductivity "K" not higher than 0.037 w/m-°K (0.26 Btu-in/hr-ft²-°F) at 24°C (75°F) mean temperature difference. Apply weatherproof treatment and provide jacket or cladding for outdoor installation.

13.2.5 Painting and Color Coding

Refrigerant pipes shall be color coded as per TES-H-107.02.

13.3 Chilled Water Piping

All insulated piping, fittings, valves, special fittings and other piping components shall be provided with outer casing (PVC, galvanized sheet) or aluminum jacket.

13.3.1 Pre-insulated Chilled Water Pipe (Underground)

Polyurethane Foam; "K" factor 0.0187 w/m-°K (0.13 Btu-in/hr-ft²-°F) at 24°C (75°F);
Over-all density 32 kg/m³ (2.0 lbs/ft³).

13.3.2 Pre-insulated Chilled Water Pipe (Above ground)

Polyurethane Foam; "K" factor 0.0187 w/m-°K (0.13 Btu-in/hr-ft²-°F) at 24°C (75°F);
Over-all density 32 kg/m³ (2.0 lbs/ft³).

13.3.3 Valves, Strainers and Special Fittings (Not Pre-insulated)

Factory molded fiberglass; density not less than 48 kg/m³ (3 lbs/ft³); conductivity "K" not higher than 0.036 w/m-K (0.25 Btu-in/hr-ft²-°F) at 37.8°C (100°F) mean temperature difference; with factory attached white vapor barrier jacket.

13.4 Buildings

13.4.1 General

Roofs of air conditioned or mechanically-ventilated buildings shall be insulated.

Floors shall be thermally insulated when the air conditioned space is above a non-conditioned space or ambient air (as in a building elevated above grade). Insulation thickness shall be determined based on optimum heat transfer coefficients.

Partitions shall be thermally insulated if the air conditioned space is adjacent to a large non-conditioned space or to an area of high heat output. Insulation thickness shall be determined based on overall coefficients of heat transfer.

13.4.2 Materials

The insulation material shall not increase the overall fire hazard classification of the assembly being insulated.

The insulation material shall be rot and rodent resistant and must not be affected by temperatures of 65°C (150°F) or by short period exposure to direct sunlight, causing thermal decomposition not related to combustion.

Insulation materials, including insulating cement and insulation covering, shall not contain asbestos fibers.

The material must be easily handled without deteriorating and without harmful effects to personnel handling it.

Where insulation is provided in a "sandwich" panel construction, thermal short-circuits (metal-to-metal contacts) shall be minimized to retain an effective insulator.

All insulation materials other than foam plastic shall have a flame-spread rating not to exceed 25 and a smoke density not to exceed 450 in accordance with UBC or equivalent SBC requirements.

Foam plastic insulation shall conform to requirements stated in UBC or equivalent SBC requirements.

Following are some of the acceptable insulation materials used for buildings as per its applications provided the advantages and disadvantages are observed accordingly.

- Non-flammable extruded polystyrene.
- Resin bonded fiberglass boards.
- Polyurethane sprayed.
- Rigid polyurethane
- Foam cement
- Non-flammable cast expanded polystyrene

To the greatest extent possible, it is necessary to take into account the following characteristics as regards to the insulation materials to be used in the exterior walls and roofs:

- Closely - celled and homogeneous in composition.
- Non-absorbent to humidity.
- High insulation capacity on the long run.
- High mechanical capability.
- Should not get hard by the time, shall be corrosion resistant to environmental circumstances prevailing in the GCC States.
- Invariable dimensions with minimal expansion or contraction.
- Highly resistant to periodical temperature fluctuation.
- Made of non-flammable material (especially where this material become easily exposed to fire.).
- Resistant to precipitation.

- Immune to fungi and non-susceptible to the growth of bacteria and microscopic organism.
- Resistant to reaction and chemical changes.

13.4.3 Design

On major projects a least cost analysis for air-conditioning requirements versus the maximum overall heat transfer coefficient [U-factor $\text{w/m}^2\text{-}^\circ\text{K}$ ($\text{Btu/ft}^2\text{-hr-}^\circ\text{F}$)] for various building types (conventional, Prefab) and building elements (stud wall, masonry concrete, wood etc.) shall be made. However, when projects are not large enough to justify detailed analysis, insulation shall be provided as necessary to achieve U-factors, no higher than those listed herein.

U-Factors, $\text{w/m}^2\text{-}^\circ\text{K}$ ($\text{Btu/hr-ft}^2\text{-}^\circ\text{F}$).

Component	Air Conditioned Building	Prefab Portable Buildings	Mechanically-Vented Buildings
Roofs	0.34 (0.06)	0.34 (0.06)	0.8 (0.14)
Walls	0.56 (0.10)	0.56 (0.10)	-
Floors	-	0.60 (0.106)	-

The design of insulated roofs and walls shall meet the criteria of ASTM C-755 for the control of water vapor flow and for the selection of vapor barriers. Vapor barriers shall be designed to be installed on the warm- side (outside) of the stud or joist. Design shall include provision for the ventilation of the insulated area to remove any accumulated vapor.

It is a must to use double layer or reflective glass in all windows of multistory buildings consisting of more than four floors (or containing more than 16 units, apartments and offices). As regards other buildings, a double layer or reflective glass shall be used where large areas of walls exceeding 10% of the total area require installation of glass.

14.0 REFRIGERANTS

Due to the confirmed reports of environmental phenomenon of Ozone Depletion and Global Warming, it has been decided to control the production and use of Chlorofluorocarbon (CFC) based refrigerants. A CFC phase-out schedule was agreed upon by the International Agencies to eliminate the use of refrigerants in order of their damage potential.

In view of this situation, the users of higher capacity air-conditioning units have the following three options:

- a. To keep operating the equipment by managing the shortage of refrigerant by methods of storage and re-cycling.
- b. To retrofit the system with a new substitute refrigerant.
- c. To dispose off the old unit and install a new model with environmentally safe refrigerant.

The decision will be based on the life of the equipment and the economics and efficiency of the system.

15.0 PIPING SYSTEMS

15.1 Hydronic Piping

Hydronic systems, used in HVAC applications, convey heat to or from a conditioned space with water as the heat transfer medium through heat transfer equipment such as heating/cooling coils, chillers, boilers, etc. These water systems can be classified according to their temperature, pressure, generation of flow, piping and pumping arrangement and can have as varied configuration as the imagination of the designer permits it. The operating success of a hydronic system is greatly affected by the many complex relationships between the various components of the system. However, regardless of system's configuration the piping circuitry shall be properly sized and combined with the pump operation to provide an efficient heat transfer system to or from the conditioned space.

15.1.1 Water Flow Rate

One of the first tasks of the HVAC Design Engineer is the determination of water flow rates in the different sections of the piping system.

These flow rates depend on the quantity of heat to be added to, or removed from the water, and the temperature difference between the entering and leaving water, in accordance with the equation of:

$$q = C_p \times M \times \Delta t$$

Where: q - heat transfer rate, kJ/s

C_p - specific heat of water, kJ/kg °C

M - mass water flow rate, kg/s

Δt - Temp. difference between entering and leaving water, °C

At 20°C water temperature, $C = 4.19$, the volumetric water flow rate can be established using the following convenient formula:

$$V = q / (4.19 \times \Delta t)$$

Where: V - volumetric water flow rate, lps

q - heat transfer rate, kW

It shall be recognized by the system designer that in large systems the rate of flow in any particular pipe header will rarely equal the sum of flow rates of all interconnected branches. In a building the individual cooling units are controlled to vary the water quantity according to the cooling load requirement of the particular area served by that unit.

Economics and good engineering practice dictates that the water piping system be designed to less than the sum of peak flow rates in each branch circuit.

In any hydronic system the water requirement and pump head pressure vary according to the changes in water demand at the individual terminals of the system.

In systems where diversity is not considered, these changes are more pronounced and may lead to excessive noise of control valves and oversized pumps, in addition to larger than required pipe header diameters.

However, excessive use of diversity may result in an undersized piping system, therefore, the use of diversity factors shall be carefully evaluated at an early stage of system design.

15.1.2 Velocity

After establishing the flow rates in the different sections of the piping system the design engineer should consider the velocity limits for pipe sizing.

Generally, designers determine preliminary pipe sizes based on velocity limitations without detailed pressure drop calculations. In hydronic systems, the following velocity limits are established on the basis of acceptable sound levels of moving water and entrained air, and on the basis of erosion limitations.

<u>Service</u>	<u>Velocity Range</u> <u>mps (fps)</u>
Chilled water, make-up water, etc.	1.22 - 3.05 (4 - 10)
Pump Discharge	2.44 - 3.66 (8 - 12)
Pump Suction	1.22 - 2.13 (4 - 7)
Drain Lines	1.22 - 22.13 (4 - 7)

These velocity limits should be used as a guide only and it shall be recognized that the final pipe sizes have to be established by detailed pressure drop calculations.

15.1.3 Pressure Drop

The general range of pipe friction loss for design of hydronic systems occurs between 1 and 4 m. head of water/100 m. length (1 and 4 ft. head of water/100 ft. length). A value of 2.5 m. head of water/100 m. length (2.5 ft. head of water/100 ft. length) represents the mean to which most systems are designed.

To determine the friction drop in a water piping system, the engineer must calculate the straight lengths of pipe sections and add the equivalent lengths of pipe due to fittings, valves and other elements in the piping system. Various tables and charts have been developed in the industry for calculating pipe friction losses and equivalent pipe lengths of fittings/valves.

When calculating system or sub-circuit pressure drops, the Designer must keep in mind that the pipe sizes have a great influence on system's first cost, operating cost and overall performance. It is, therefore, of prime importance that head loss calculations be carried out in detail and the values arrived at shall not be inflated by the application of various safety factors.

15.1.4 System Characteristics Curve

On any hydronic system the head loss varies at the square of the flow as expressed by the following equation:

$$H_2 / H_1 = (Q_2 / Q_1)^2$$

Where: H_1 = Head Loss at initial condition, kPa (or ft. wg.)

H_2 = Head Loss at final condition, kPa (or ft. wg.)

Q_1 = Flow Rate at initial condition, lps (or gpm)

Q_2 = Flow Rate at final condition, lps (or gpm)

From this relationship the system characteristics curve, indicating system head requirements at various flow rates, can be plotted.

The application of this head/flow relationship is basic to piping system design, pump selection, balancing and trouble shooting.

15.1.5 Pump Selection

Pump selection for hydronic systems must be accomplished through careful matching of the pump operating characteristics to the system's operating requirements.

Performance characteristics of centrifugal pumps are indicated by pump curves that plot flow rate against pump Total Dynamic Head (TDH) along with other information such as efficiency, brake horsepower and Net Positive Suction Head (NPSH) requirement.

Centrifugal pumps are designed around their Best Efficiency Points (BEP) where they have the best operating condition at the most economical energy input level.

While the hydraulic losses are at their minimum at the BEP, the least efficient regions of operation are around zero flow and at the end of the curve in the maximum flow area.

Although the preferred range for centrifugal pump selection is around the BEP, pump performance shall be considered across the entire characteristics curve.

Centrifugal pumps are available with steep performance curves with large difference in TDH between zero flow and maximum flow conditions, and with flat curves that show only a minimum TDH difference between shut-off and maximum flow conditions.

For a hydronic system with constant flow rate and relatively high pressure drop across branch circuits the use of a steep pump curve is preferable. In case the actual system curve is different from the calculated curve, the steep pump curve will intersect the system curve closer to the designed operating point.

For a system with variable flow rate at the terminal units, however, a pump with flat performance curve is preferred. In such systems a change in flow rate at any terminal unit produces only a minimum change in pump head requirement and does not disturb the balance of the whole system.

The required electric motor sizes are indicated on the pump performance charts, but the true power requirement can be checked at any point on a pump performance curve as follows:

- a. Water Power in English Units:
 Water power in Hp = $\frac{Q \times \text{TDH} \times \text{SG}}{3960}$
- b. Water Power in SI (Metric) Units:
 Water power in kW = $0.0098 \times Q \times \text{TDH} \times \text{SG}$

$$= \frac{Q \times \text{TDH (kPa)} \times \text{SG}}{102}$$

Brake power in kW = Water Power/Pump Efficiency

Where: TDH = Total Dynamic Head, in m.

Q = Flow Rate, lps (or gpm)

SG = specific gravity of water = 1

From a known operating condition pump performance can be calculated for a new condition according to the relationship of affinity laws or pump laws:

- Flow (capacity) varies directly as the speed or impeller diameter
- Head varies as the square of the speed or impeller diameter
- Brake power varies as the cube of the speed or impeller diameter.

Based on affinity laws, pump equation can be derived as follows:

- a. With impeller diameter, D, held constant:

1. $Q_2/Q_1 = N_2/N_1$
2. $H_2/H_1 = (N_2/N_1)^2$
3. $BP_2/BP_1 = (N_2/N_1)^3$

- b. With speed, N, held constant:

1. $Q_2/Q_1 = D_2/D_1$
2. $H_2/H_1 = (D_2/D_1)^2$
3. $BP_2/BP_1 = (D_2/D_1)^3$

Where:

- Q_2 - new or required water flow, lps (or gpm)
- Q_1 - actual or existing water flow, lps (or gpm)
- H_2 - new or required head, kPa (or ft. wg.)
- H_1 - actual or existing head, kPa (or ft. wg.)
- BP_2 - new or required brake power, kW (or Bhp)
- BP_1 - actual or existing brake power, kW (or Bhp)
- D_2 - new or required impeller diameter, mm. (or in.)
- D_1 - actual or existing impeller diameter, mm. (or in.)
- N_2 - new or required pump speed, Rad/sec (or RPM)
- N_1 - actual or existing pump speed, Rad/sec (or RPM)

Particular attention must be given to the pump suction pressure requirement, especially when handling hot liquid. If the pump suction pressure approaches the vapor pressure of the liquid, vapor pockets will form in the impeller passages. The collapse of the vapor pockets (cavitations) will cause excessive noise, reduced performance and premature pump failure.

The required pump suction pressure to prevent cavitation is expressed in terms of Net Positive Suction Head (NPSH), and the available NPSH can be calculated as follows:

$$\text{Available NPSH} = H_a + H_z - H_f - H_{vp}$$

Where: NPSH = Net Positive Suction Head, Pa (or ft. wg.)

H_a = Absolute pressure on liquid surface where pump takes suction, Pa (or ft. wg.)

H_z = Elevation head (static head) above or below the pump center line. If above, positive static head. If below, negative static head sometimes referred to as suction lift, Pa (or ft. wg.)

H_f = Friction head on the suction side of the system including piping, fittings, valves, heat exchanges at the design velocity within suction system, Pa (or ft wg.)

H_{vp} = Absolute vapor pressure at pumping temperature, Pa (or ft. wg.)

For a trouble-free operation, the available NPSH always has to be larger than the required NPSH.

HYDRONIC PIPING DESIGN DATA

NODE NO. (From/To)	FLOW RATE (lps)	PIPE SIZE (mm)	VELOCITY (mps)	FITTING TYPE AND NUMBER	PIPE LENGTH (m)	FITTING EQV. LENGTH (m)	TOTAL EQV. LENGTH (m)	FRICTION LOSS Pa (head)/ m. (length)	FRICTION HEAD (Pa)

15.2 Refrigerant Piping

The design of refrigerant piping systems requires some special considerations in addition to the general principles involved in the design of other fluid flow systems. These special considerations include the following:

- Proper refrigerant feed to the evaporator must be assured by designing the system so that the liquid refrigerant entering the metering device has adequate pressure and is free of gas.
- Prevention of trapping lubricating oil in the system by maintaining the required minimum oil-carrying velocities even at partial load conditions.

Refrigerant flow rates required for different system capacities are indicated in ASHRAE Fundamentals, and practical refrigerant gas line velocities for the different type of refrigerants are given in the same handbook.

To eliminate problems from liquid hammering due to operation of solenoid valves, liquid line velocity shall be sized at or under 1.52 mps (300 FPM).

For most efficient compressor capacity and low equipment operating costs, suction line velocity shall be designed at 4.57 - 20.32 mps (900 - 4000 FPM).

However, maximum system pressure drop and minimum line velocity requirements for oil return may override these velocity recommendations.

Pressure drop limitations for liquid, suction and discharge (hot gas) lines are described in Sections 6.1.1 to 6.1.3. The limitations for good oil return are discussed in ASHRAE Systems and Equipment.

The following is a recommended procedure for sizing refrigerant piping:

- a. Measure the total length of straight pipes.
- b. Add 50% to obtain preliminary total equivalent length
- c. Read pipe size from refrigerant pipe sizing charts ASHRAE RP185.
- d. Calculate equivalent length of valves and fittings and add it to pipe length to find total equivalent length.
- e. Check pipe sizes against total equivalent length in refrigerant pipe sizing charts.
- f. Repeat steps d) and e) if necessary.

If the actual pressure drop has to be found in the refrigerant lines, the line sizing procedure shall be continued as follows:

- g. Find the volumetric flow rates in the lines using density values from refrigerant tables.
- h. Calculate friction losses in pipes.
- i. Determine pressure drop through valves and accessories from vendors' catalogs.
- j. Calculate static head requirements
- k. Summarize the results of steps (h), (i) and (j).
- l. If the total pressure drop from step (k) is too high, change pipe size and repeat steps (h) through (l).

15.2.1 Liquid Lines

Liquid line pressure drop shall not be excessive to avoid gas formation or insufficient pressure at the liquid feed device. Liquid lines are normally sized for a maximum friction loss that corresponds to about 0.6 - 1.2°C (1 - 2°F) change in saturation temperature.

Liquid pressure losses for a change of 0.6°C (1°F) saturation at 55°C (100°F) condensing pressure are approximately:

<u>Refrigerant</u>	<u>Change, kPa (psi)</u>
R-134a	14.5 (2.1)

Oil return in liquid lines usually does not present a problem, since the refrigerant oil is adequately miscible with halocarbons in the liquid form.

However, it is important that the liquid refrigerant entering the liquid metering device is sub-cooled and gas-free.

15.2.2 Suction Lines

Suction line pressure drop decreases system capacity by lowering compressor suction pressure.

Normally, suction lines are sized for a maximum friction loss of 1.2°C (2°F) change in saturation temperature.

At 4.4°C (40°F) saturated temperature at the compressor suction the following are the maximum recommended suction line friction losses:

<u>Refrigerant</u>	<u>Change, kPa (psi)</u>
R-134a	13.31 (1.93)

Below 4.4°C (40°F) compressor suction pressure, the pressure drop equivalent of any temperature change decreases. For this reason, in very low temperature applications liquid lines have to be sized for lower pressure drops than the above table indicates. However, the suction line risers must maintain the minimum required velocity for effective oil return to the compressor.

If the diameter of suction riser has to be reduced for the sake of effective oil return, the additional friction loss of the riser has to be compensated by oversized horizontal and down coming pipe sections.

15.2.3 Discharge Lines

Pressure drop in discharge lines decreases compressor capacity and shall be kept to a minimum, but the line velocity shall be adequately high for carrying the entrained oil. For most applications both of the above criteria can be satisfied with a single hot gas (discharge) riser. However, in some cases of multiple compressor installation the hot gas riser sized to entrain oil at minimum load has an excessive pressure drop at maximum load. In such a case double risers can be used or an oil separator has to be installed in the discharge line just ahead of the single riser.

REFRIGERANT PIPING DESIGN DATA CHART

	LIQUID	SUCTION	DISCHARGE
System Capacity, kW			
Flow Rate, kg/min.			
Condensing Temperature, °C			
Saturated Suction Temperature, °C			
Density, kg/m ³			
Volumetric Flow, m ³ /min.			
Line Velocity, m/min.			
Pipe Length, m			
Valves and Fittings Equivalent Length, m			
Total Equivalent Length, m			
Pipe Diameter, mm			
Friction Loss, m. head/100 m length			
Head Loss, m			

15.3 Material Selection

15.3.1 Pre-insulated Chilled Water (Underground):

- Carrier Pipe: Steel Sch. 40 or 80; ASTM A120, ASTM A53, ASTM A333, API 5L; Seamless
- Insulation : Polyurethane Foam, "K" factor 0.0187 w/m-°K (0.13 Btu-in/hr-ft²-°F) at 24°C (75°F), overall density 32 kg/m³ (2.0 lbs/ft³)
- Casing: PVC Type I Grade I, ASTM D1784

15.3.2 Chilled Water (Above Ground):

- Carrier Pipe: Steel Sch. 40 or 80; ASTM A120, ASTM A53, ASTM A333, API 5L; Seamless
- Insulation : Polyurethane Foam, "K" factor 0.0187 w/m-°K (0.13 Btu-in/hr-ft²-°F) at 24°C (75°F), overall density 32 kg/m³ (2.0 lbs/ft³)
- Casing: Spirally locked seamed galvanized sheet metal having a minimum wall thickness of 0.55 mm.

15.3.3 Refrigeration Piping:

- Pipe: Nitrogenized ACR Type Hard Drawn Copper Tube (ASTM B-280)
- Fittings: Wrought Copper (ANSI B16.22)
- Solder: Silver solder alloy; on valves, use solder as recommended by the manufacturer.

16.0 FRESH AIR MAKE-UP AND MECHANICAL ROOM

16.1 Fresh Outdoor Air Intake

- Fresh air intakes shall be constructed and located to minimize dust intrusion. The openings shall be provided with removable screens fabricated of gauge 16 galvanized steel, 13 mm (1/2 in.) wire mesh with a galvanized steel framing.
- The fresh outdoor air make-up system shall consist of weatherproof louver, insect/bird screen, sand trap, volume damper, final filter and ductwork.
- The fresh air intake system may also consist of weatherproof louver, insect/bird screen, volume damper, pre-filter (in lieu of sand trap), final filter and ductwork.
- Pre-filter and final filter dust spot efficiency and weight arrestance rating shall be in accordance with Section 12.
- Intake louver sizing shall be per ASHRAE Fundamental Handbook.

16.2 Mechanical Room

The mechanical room shall be sized to provide 900 mm (3 ft.) minimum clearance as on each side of equipment with operable panels or controls, unless a greater clearance is required by equipment manufacturer. Adequate space shall be provided for coil removal. (Normally, access for coil removal should be sized 1.20 times the width of AHU).

17.0 SUPERVISORY CONTROL SYSTEM

17.1 Introduction

The increasing size of modern buildings and building complexes, and the difficulty of obtaining competent operating personnel have led to an increasing use of a computer-based central supervisory control system. These systems allow one person, at a central location, to monitor and control the operation of up to several thousand elements of the HVAC & R systems in the facility. The addition of alarm circuits and audio-visual communications improves security as well as simplifying and improving the degree of control which can be maintained. The additional cost of such a system can be amortized in a few years by more efficient operation and maintenance. Where numerous tenants are involved, the increased tenant satisfaction may be sufficient justification for the added cost.

17.2 Computer-Based Systems for Monitoring and Control

The term “computer-based” refers to a control system which utilizes programmable digital computer. These computer-based systems are called by various names: MCS (Monitoring and Control System), BMS (Building Management System), BAS (Building Automation System), EMCS (Energy Management and Control System), EMS (Energy Management System) and others. Functionally, they are all equivalent.

The primary function of a computer-based system is to provide monitoring, intervention control and direct control of local loop elements (usually called Direct Digital Control). It can also provide historical data summaries, data analysis and maintenance scheduling.

Security and fire reporting functions are fundamentally the same monitor-control-alarm functions used in HVAC supervision. The systems which are used for HVAC control can therefore easily be adapted to these additional functions.

17.3 Industry Specifications and Systems Advantages

The advantages of having supervisory control systems are programmability, energy conservation by precise control of temperature and equipment scheduling, remote accessibility, prevention of unauthorized tampering and centralization.

18.0 ENERGY CONSERVATION

18.1 General

The air conditioning and ventilation equipment consumes a major portion of the electricity generated in the Kingdom of Saudi Arabia. Therefore, the emphasis should be to design a system for new or retrofit projects that will have the lowest energy consumption over the operating life of the facility while meeting the user's need.

18.2 Factors

The factors affecting energy conservation can be listed as follows:

- HVAC Design: Attention should be given to U-values for walls, floors, roofs, glass and the zoning of spaces. Also, ventilation air requirement should be carefully selected without compromising on the Indoor Air Quality (IAQ) to avoid Sick Building Syndrome (SBS).
- HVAC Equipment Selection: It should be proper, with products having higher Energy Efficiency Ratio (EER) and advanced features of microprocessor technology to control and monitor the various design parameters in an efficient manner.
- HVAC Testing, Adjusting and Balancing: It should be properly performed by a firm who specialize in this field.
- O & M: Proper operation scheduling to meet load requirement and periodic equipment maintenance.

19.0 DESIGN DOCUMENTATION

19.1 Specifications

Design Execution Specifications and Design Drawings shall be prepared for each project. These Specifications shall include all mechanical design requirements for Heating, Ventilation and Air-Conditioning (HVAC). All applicable National Grid Saudi Arabia Engineering Standard requirements shall be covered in the design documentation.

Specifications shall be complete and include work description, references to drawings and standards, and description of construction materials.

19.2 Drawings

HVAC drawings shall be provided with sufficient details to permit the construction of a complete facility. Drawings shall include a list of drafting symbols/abbreviations used, and equipment schedules.

The HVAC system design criteria shall be indicated on the design drawings and shall include, but not limited to, the following data:

- a. Outside air conditions used for load calculations;
- b. Indoor design conditions;
- c. Total sensible cooling load with a breakdown to its components, i.e. external heat gain, heat from lights, people, etc.
- d. Total latent cooling load with a breakdown;
- e. Grand total cooling load;
- f. Heating load;
- g. Outside air intake;
- h. Chilled water (if any) flow rate, temperature rise and pressure drop;
- i. Entering and leaving air conditions at cooling coil; and
- j. Total pressure drop of air handling system with a breakdown to components, i.e. pressure drop through supply/return ducting, coils, filters, etc.

19.3 Equipment Schedules

Equipment schedules shall be included on the drawings with information necessary for bidding, purchasing, and installation of all equipment. Schedules shall contain minimum required data as shown on the attached forms for the following:

- a. Air Handling Unit (Direct Expansion Type)
- b. Air Handling Unit (Chilled Water Type)
- c. Air-Cooled Condensing Unit
- d. Air Filter
- e. Packaged Chiller
- f. Pump
- g. Fan
- h. Electric Duct Heater
- i. Air Terminal

COOLING LOAD TEMPERATURE DIFFERENCE (CORRECTED)

NATIONAL GRID SAUDI ARABIA FORM NO. 16307 (07/10)

				①	②	③	④	⑤	⑥
ITEM NO.	ITEM	DIRECTION	CLTD FROM TABLE 16 HOURS	JUST FOR LATITUDE & MONTH	COLOR CORR. "K"	CORR. INDOOR TEMP (78-Trm)	CORR. OUTDOOR TEMP (Avg. Toa-85)	CLTD CORR.	
1	ROOF								
2	WALL								
3	WALL								
4	WALL								
5	WALL								
6	GLASS								
7	SKYLIGHT								

① From Architectural Plans

② For Roofs: See ASHRAE Fundamentals.

For Walls: See ASHRAE Fundamentals.

For Glass: See ASHRAE Fundamentals

③ For Roofs: See ASHRAE GRP 158 – for horizontal surface

For Walls: See ASHRAE GRP 158

④ "K" is a color correction to be applied after the month latitude adjustment.

 Roofs: K = 1.0 Dark color
 K = 0.5 Light color

 Walls: k = 1.0 Dark color
 k = 0.83 Medium colors
 K = 0.75 Light colors

⑤ (78 – Trm) = Room air design temperature correction

Trm = Room air design temperature °F

⑥ (Ave. Toa – 85) = Outdoor air design temperature correction

Ave. Toa = Average outside air temperature on design day, °F

Ave. Toa = Toa – 0.5 daily range

ROOM COOLING LOAD CALCULATION

ROOM NO. _____		AREA ft	U FACTOR	BASE CLTD	CORR. CLTD	SC	SHFG	CLF	HEAT GAIN, Btu/hr	
									SENS.	LAT.
ROOF										
		2								
EXTERIOR WALLS AND DOORS	N					-	-	-		-
	NE					-	-	-		-
	E					-	-	-		-
	SE					-	-	-		-
	S					-	-	-		-
	SW					-	-	-		-
	W					-	-	-		-
	NW					-	-	-		-
EXTERIOR GLASS	COND	--								-
	S	N		-	-	-				-
		NNE / NNW		-	-	-				-
	O	NE / NW		-	-	-				-
		ENE / WNW		-	-	-				-
	L	E / W		-	-	-				-
		ESE / WSW		-	-	-				-
A	SE / SW		-	-	-				-	
	SSE / SSW		-	-	-				-	
R	S		-	-	-				-	
	HORIZ.		-	-	-				-	
PARTITION/CEILING/FLOOR					Delta t = _____ °F			-		-
TOTAL EXTERNAL HEAT GAIN										-
LIGHTS		INPUT RATING x CLF								-
PEOPLE	SENSIBLE	NO. OF PEOPLE x SENSIBLE HEAT GAIN x CLF								-
	LATENT	NO. OF PEOPLE x LATENT HEAT GAIN							-	
APPLIANCES	SENSIBLE	HEAT GAIN x CLF								-
	LATENT	HEAT GAIN							-	
POWER		HEAT GAIN x CLF								-
VENTILATION &	SENSIBLE	1.10 x CFM _{oa} x (t _{oa} - t _{rm})								-
INFILTRATION	LATENT	0.70 x CFM _{oa} x (W _{oa} - W _{rm})							-	
TOTAL ROOM HEAT GAIN										
SUPPLY AIR REQUIRED =		TOTAL ROOM SENSIBLE HEAT GAIN					=		CFM	
		$1.10 \times (t_{rm} - t_{sa})$								

rcl.xls

Form No. 16305(07/10)

COOLING & HEATING LOAD ESTIMATE SHEET

DESIGN CONDITIONS	SUMMER		OUTDOOR	ROOM	DELTA T
	A	DRY BULB T	° F	° F	° F
	B	WETBULB	° F	° F	
	C	HUMIDITY RATIO (W)	W _{oa}	W _{in}	
	WINTER		OUTDOOR	ROOM	DELTA T
D	DRY BULB	° F	° F	° F	
E	HUMIDITY RATIO, W*	W _{oa}	W _{in}		

LATITUDE	
DESIGN MONTH	
DESIGN HOUR	
DAILY RANGE	° F

PROJECT			
LOCATION			
BLDGRM. NO.			
LENGTH		WIDTH	
HEIGHT		VOLUME	

SHEET NO.	OF
DATE	
CONTRACT NO.	
B.I. NO.	
J.O. NO.	
ESTIMATED BY	
CHECKED BY	

*GRAINS OF MOISTURE PER LB. DRY AIR (W_{oa} - W_{in})**GRAINS OF MOISTURE PER LB. DRY AIR (W_{in} - W_{oa})

EXTERNAL COOLING LOAD	ITEM NO.	ITEM	DIRECTION	CLTD CORR. x	U VALUE x	AREA SQ. FT.
	1	ROOF				
	2	WALL				
	3	WALL				
	4	WALL				
	5	WALL				
	6	GLASS				
	7	SKYLIGHT				

INTERNAL COOLING LOAD	ITEM NO.	ITEM	U FACTOR x	AREA SQ. FT.	DELTA T
	8	PARTITION WALL			
	9	CEILING			
	10	FLOOR			
	11				

SOLAR	ITEM NO.	ITEM	DIRECTION	SHADE COEFF (SC) x	SHCF x	AREA SQ. FT.	CLF
	12	GLASS	UNSHADED				
			SHADED				
	13	GLASS	UNSHADED				
			SHADED				
	14	GLASS	UNSHADED				
		SHADED					
15	SKYLIGHT	UNSHADED					
		SHADED					

LIGHTS	ITEM NO.	TYPE	TOTAL HOURS ON	TOTAL WATTS x	WATTS TO BULB x	USAGE % x	BALLAST FACTOR x	CLF
	16	FLUORESCENT			341			
	17	INCANDESCENT			341			

PEOPLE	ITEM NO.	NO. OF PEOPLE x	q _s PER PERSON x	CLF
	18			
	19			

APPLIANCE	ITEM NO.	HOODED YES - NO	BULB/ry SENS. q _s x	CLF
	20			
	21			
	22			
	23			
	24			

EQUIP	ITEM NO.	ADDITIONAL HEAT GAINS
	25	
	26	

27	ROOM SENSIBLE SUBTOTAL	
28	SAFETY FACTOR (R ₉₀) x ITEM 27	
29	ROOM SENSIBLE HEAT (ITEMS 27 + 28)	

30	VENTILATION LOAD	CFM x 1.10 x DELTA T (db)
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LATENT LOAD	ITEM	q LATENT x	NUMBER
	31	PEOPLE	
	32	EQUIPMENT	
	33	EQUIPMENT	
	34	ROOM LATENT	
	35	VENTILATION LOAD	CFM x 0.70 x (W _{oa} - W _{in})

COOLING

1 SENSIBLE Bulb/ry

HEATING

U VALUE	AREA SQ. FT.	q SENSIBLE Bulb/ry

SUB TOTAL		
VENTILATION AIR		
1.1 x DELTA T x CFM		

TO TOTAL PICK-UP TOTAL		
GRAND TOTAL		

HUMIDIFICATION		
0.7 x DELTA W** x CFM		

** GRAINS OF MOISTURE PER LB OF DRY AIR, (W_{in} - W_{oa})

COOLING LOAD SUMMARY

SENSIBLE HEAT		
36	ROOM SENSIBLE (ITEM 29)	
37	VENTILATION AIR SENSIBLE (ITEM 30)	
38	RAN MOTOR HEAT GAIN	
39	GRAND SENSIBLE HEAT (ITEMS 36 + 37 + 38)	
44	TO TALE EQUIPMENT LOAD (ITEMS 39 + 40)	
45	WONING EQUIVALENT OF EQUIPMENT LOAD	ITEM 44 12000

LATENT HEAT		
40	ROOM LATENT (ITEM 34)	
41	VENTILATION AIR LATENT (ITEM 35)	
42	GRAND LATENT ITEMS (ITEM 40 + 41)	
46	ROOM SENSIBLE HEAT FACTOR	ITEM 36 ITEMS 36 + 40
47	CFM SUPPLY AIR	ITEM 36 1.1 x (T _{in} - T _{db})
48	GRAND SENSIBLE HEAT FACTOR	ITEM 39 ITEMS 39 + 40

(A) AIR HANDLING UNIT SCHEDULE (DX TYPE)*

DESIGNATION	SERVICE/ LOCATION	TYPE	FAN SECTION						
			SUPPLY AIR VOL. CUM	OUTSIDE AIR VOL. CUM	TSP, PA	ESP, PA	FAN RPM	MOTOR KW	V/PH/HZ

COOLING COIL SECTION								
COOLING CAPACITY		FACE VEL. mps	ROWS/ FINS PER CM	AIR PRESS DROP, Pa	ENTERING AIR TEMP °C		LEAVING AIR TEMP °C	
TOTAL KW	SENSIBLE KW				db	wb		

HEATING COIL SECTION				REMARKS
HEATING CAP, KW	NO. OF STAGES	VOLT/PH	AMPS	

* APPLICABLE FORM ALSO FOR PACKAGE/ROOF TOP UNIT

(B) AIR HANDLING UNIT SCHEDULE (CHILLED WATER TYPE)

DESIGNATION	SERVICE / LOCATION	TYPE	FAN SECTION						
			SUPPLY AIR VOL., CMH	OUTSIDE AIR VOL., CMH	TSP, Pa	ESP, Pa	FAN RPM	MOTOR kW	V / PH / HZ

COOLING COIL SECTION cont'd.								
COOLING CAPACITY		FACE VEL. mps	ROWS / FINS PER CM.	AIR PRESS. DROP, Pa	ENTERING AIR TEMP., C°		LEAVING AIR TEMP., C°	
TOTAL, kW	SENSIBLE, kW				db	wb	db	wb

cont'n COOLING COIL SECTION					REMARK
WATER FLOW RATE, lps	WATER VEL. mps	WATER PD, kPa	ENT. WATER TEMP., C°	LVG. WATER TEMP., C°	

(C) AIR-COOLED CONDENSING UNIT SCHEDULE

DESIGNATION	SERVICE / LOCATION	COOLING CAP., kW	UNIT INPUT, kW	NO. OF REF. CIRCUITS	NO. OF COMP.	CAP. CONTROL STEPS

AMBIENT TEMP., °C	SAT. SUCTION TEMP., °C	SAT. COND. TEMP., °C	POWER SUPPLY V / PH / HZ	REMARKS

(D) AIR FILTER SCHEDULE

DESIGNATION	SERVICE / LOCATION	TYPE	NUMBER / SIZE	PRESSURE DROP, Pa		REMARKS
				CLEAN	DIRTY	

(E) PACKAGED CHILLER

DESIG-NATION	SERVICE / LOCATION	TYPE	COOLING CAP., kW	UNIT INPUT kW	EVAPORATOR					COND. AMBIENT TEMP., °C	REMARKS
					WATER	lps	PD, kPa	EW.T, °C	LWT, °C	FOULING	

(F) PUMP

DESIG-NATION	SERVICE / LOCATION	TYPE	WATER FLOW, lps	TDH kPa	REQ. NPSH kPa	PUMP RPM	MOTOR		REMARKS
							kW (Hp)	V / PH / HZ	

(G) FAN SCHEDULE

DESIG-NATION	SERVICE / LOCATION	TYPE / CLASS	FAN				MOTOR		REMARKS
			CAP., CMH	TSP, Pa	DRIVE	RPM	kW (HP)	V / PH / HZ	

(H) ELECTRIC DUCT HEATER SCHEDULE

DESIG-NATION	SERVICE / LOCATION	HEATER CAP., kW	NO. OF STAGES	VOLT / PH/HZ	AIR FLOW CMH	FACE AREA SQ.M.	MIN. AIR VEL., mps	REMARKS

(I) AIR TERMINAL SCHEDULE

SUPPLY AIR OUTLET							
ROOM SERVED	DESIG-NATION	OVER-ALL SIZE, mm.	AIR VOL. CMH	TYPE	QTY.	NC RATING MAX.	REMARKS
RETURN AIR INLET							
TRANSFER AIR OPENING							